



FABRIC MANUFACTURING II

Part: B

Course Code: FE 351-0723

Warp Knitting Machineries

Arnob Mahmud

Lecturer

Department of Fabric Engineering

Bangladesh University of Textiles (BUTEX)

E-mail: arnob@fe.butex.edu.bd

CLASSES OF WARP KNITTING MACHINES

3 Main warp knitting machines:

1. TRICOT Warp Knitting Machine
 - Use Bearded & Compound Needles.
 - Fine gauge needles.
2. RASCHEL Warp Knitting Machine
 - Use Latch Needles.
 - Relatively coarse gauge needles.
3. CROCHET Warp Knitting Machine
 - Use Latch needles.

CLASSES OF WARP KNITTING MACHINES

1. Based on the number of needle bars

- A. One needle bar (single needle bar machines): Only plain knitted structures can be produced. Fabrics are called single face fabrics.
- B. Two needle bars (double bar warp knitting machines): Rib knitted structures can be produced. Fabrics are called double face fabrics. Needles should be in rib gaiting.

2. Depending on the type of needles employed

- A. Single needle bar machines: Tricot machines (Bearded needles), Raschel machines (Latch needles)
- B. Double needle bar machines: Simplex machines (Bearded needles), Raschel machines (Latch needles)

CLASSES OF WARP KNITTING MACHINES

3. Depending on the Drive mechanism

- A. Cams: In older machines; slow running machines. Reliable up to about 500 rpm
- B. Eccentrics and crank systems: Modern high-speed machines. Speeds up to 3500 RPM possible. Today, compound needles have replaced bearded and latch needles.

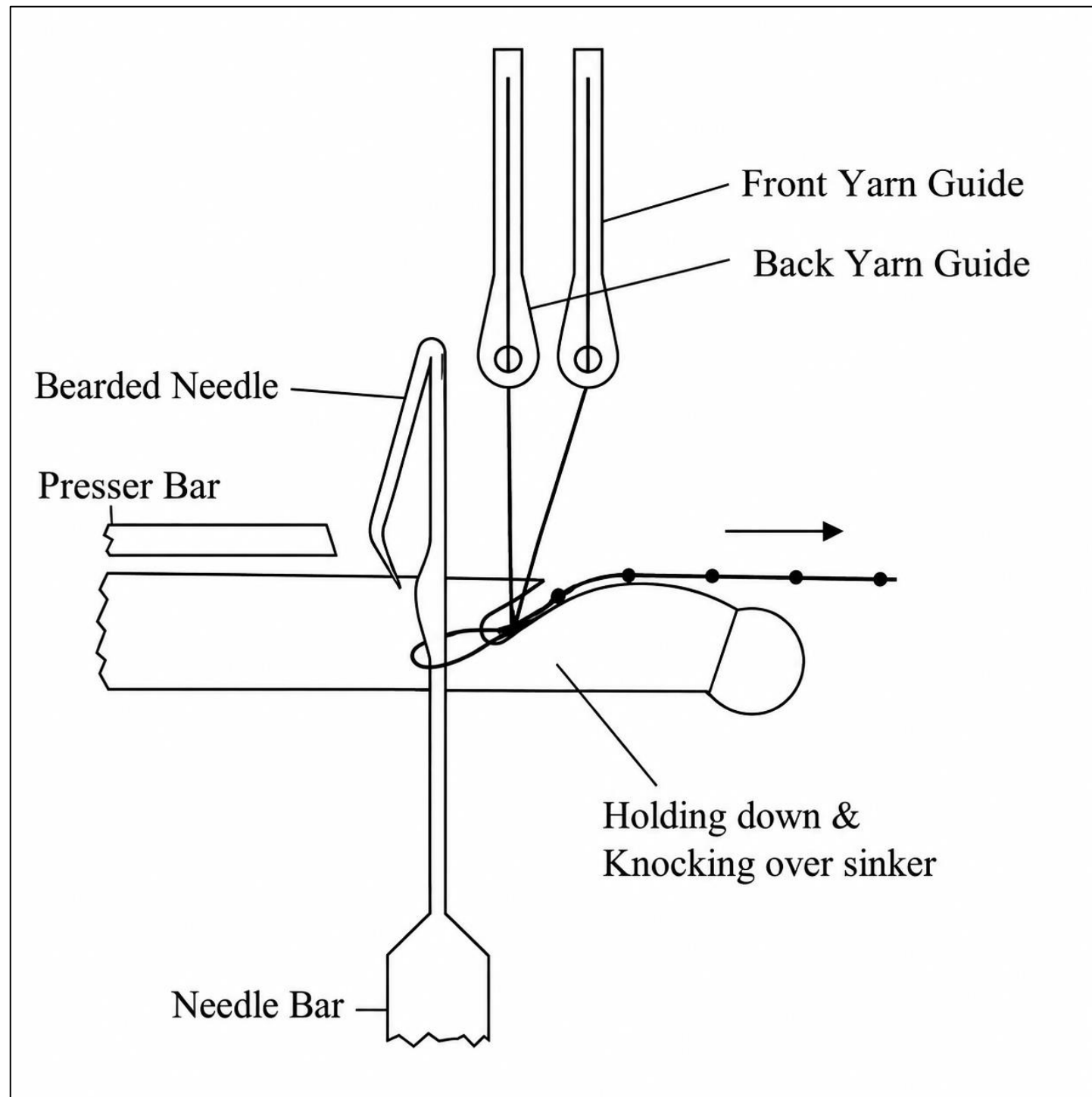
4. Classification is based on the Function of the Sinkers used

- A. Holding-down and knock-over sinkers (Tricot Machines): Holding-down of the knitted loops & provision of a platform for interconnection of knitted loops.
- B. Holding-down sinkers (Raschel Machines): Holding-down of the knitted loops.

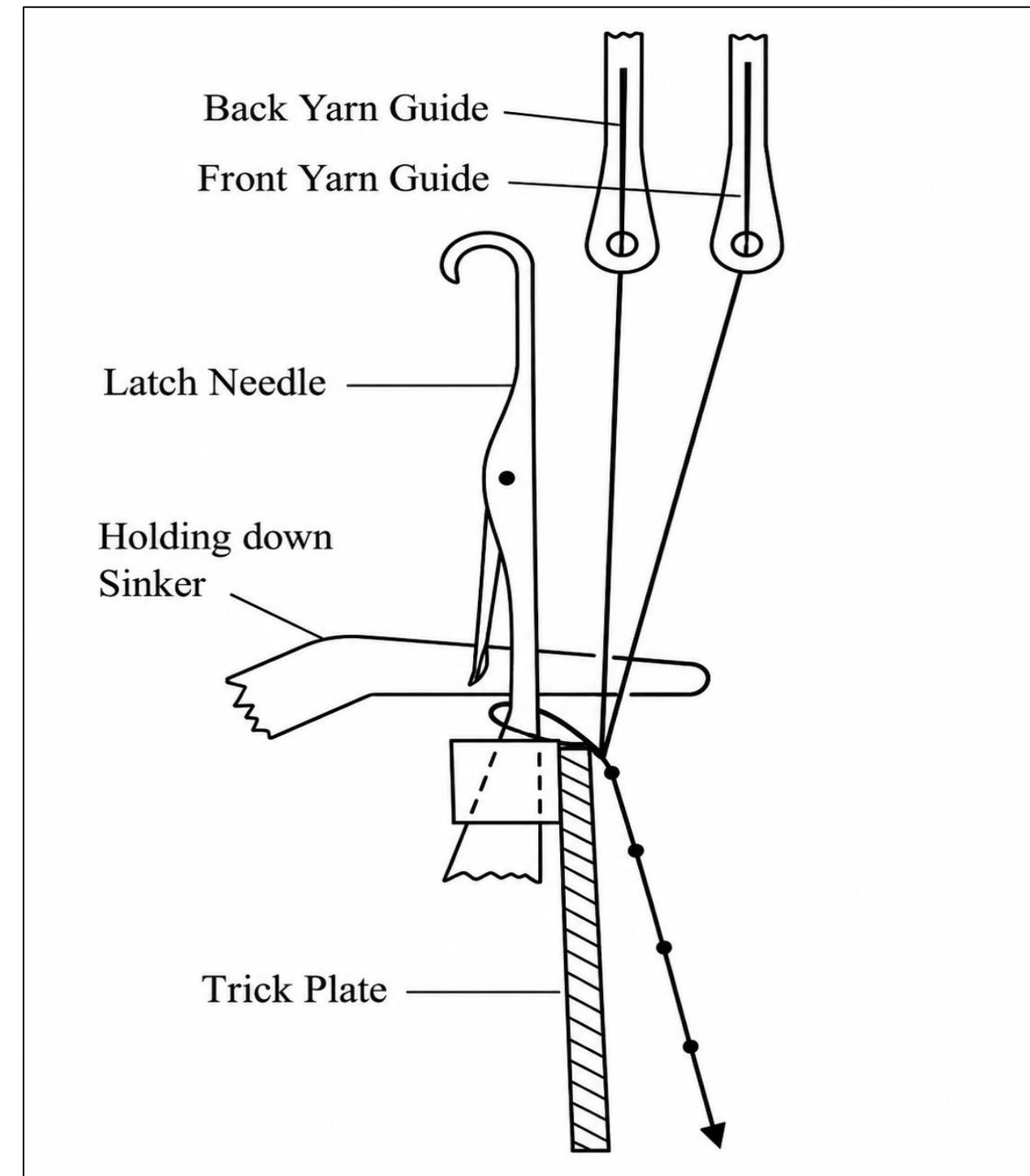
5. Direction of Fabric Removal from the needles

- A. In Tricot Machines: At right angle to the needle stems.
- B. In Raschel Machines: Almost parallel to the needle stems (120 to 160 degrees).

WARP KNITTING MACHINES



TRICOT WARP KNITTING



RASCHEL WARP KNITTING

FEATURES OF TRICOT WARP KNITTING MACHINE

1. Type of needle: Tricot m/cs mainly employed bearded needles with a pressure bar. But now, almost all available commercial tricot fabric produced by compound needle.
2. Machine gauge: Tricot m/cs have a gauge expressed in needle per inch.
3. Chain link numbering: Chain link numbering 0, 1, 2, 3, 4 etc. Generally, with three links per course.
4. Tricot sinker: Their sinkers, which are joined to each other at the front and back, never move clear of the needles as they combine the functions of holding down, knocking over and supporting the fabric
5. The Fabric is drawn-away towards the batching roller almost at right angles to the needles bar.
6. The warp beams are accommodated in an inclined arc towards the back of the m/c with the top beam supplying the front guide bar and the bottom beam supplying the back guide bar.
7. The warp sheets pass over the top of the guide bar rocker shaft to their tension rails situated at the front of the m/c.
8. Mechanical attention to the knitting elements is carried out at the front of the m/c, as the beams prevent access to the back.

FEATURES OF TRICOT WARP KNITTING MACHINE

9. As all the warp sheets are drawn over the rocker shaft to the m/c, it is easier to thread up the guide bars commencing with the needle bar.
10. The guide bars are therefore numbered from the back towards the front of the m/c because of threading sequence.



FUNCTION OF VARIOUS PARTS OF TRICOT WARP KNITTING MACHINE

1. Compound needle: In Tricot warp knitting m/c **compound needle** is used. To form loop and produce the fabric is the main function of the needle.
2. Needle bar: A needle bar is used in this m/c. The main function of it is to **hold the needles together** and helps the needles to move unison while loop forming.
3. Guide bar: Guide bar is used in this m/c to **guide the yarn properly to the needle**. It feeds the yarn around the needle and controls the rate of warp feed from the warp beam by making lapping movement. Function of different parts of tricot machine.
4. Sinker and Sinker bar: In the tricot warp knitting m/c **tricot sinker is used to hold down the loops produced by the needles**. The sinker bar keeps the sinkers together to move unison while knitting. The main function of the sinker is to hold down, knock over and supporting the fabric loops.
5. Sliding latch: In warp knitting m/c compound needles are used. A sliding latch is used here **to close the hook while knitting**.
6. Sliding latch bar: In warp knitting m/c the sliding latches remain unison. **The latch bar keeps the latches together and helps it to move unison while knitting**.
7. Cloth roller: The produced **fabric is wound** on the cloth roller.

FUNCTION OF VARIOUS PARTS OF TRICOT WARP KNITTING MACHINE

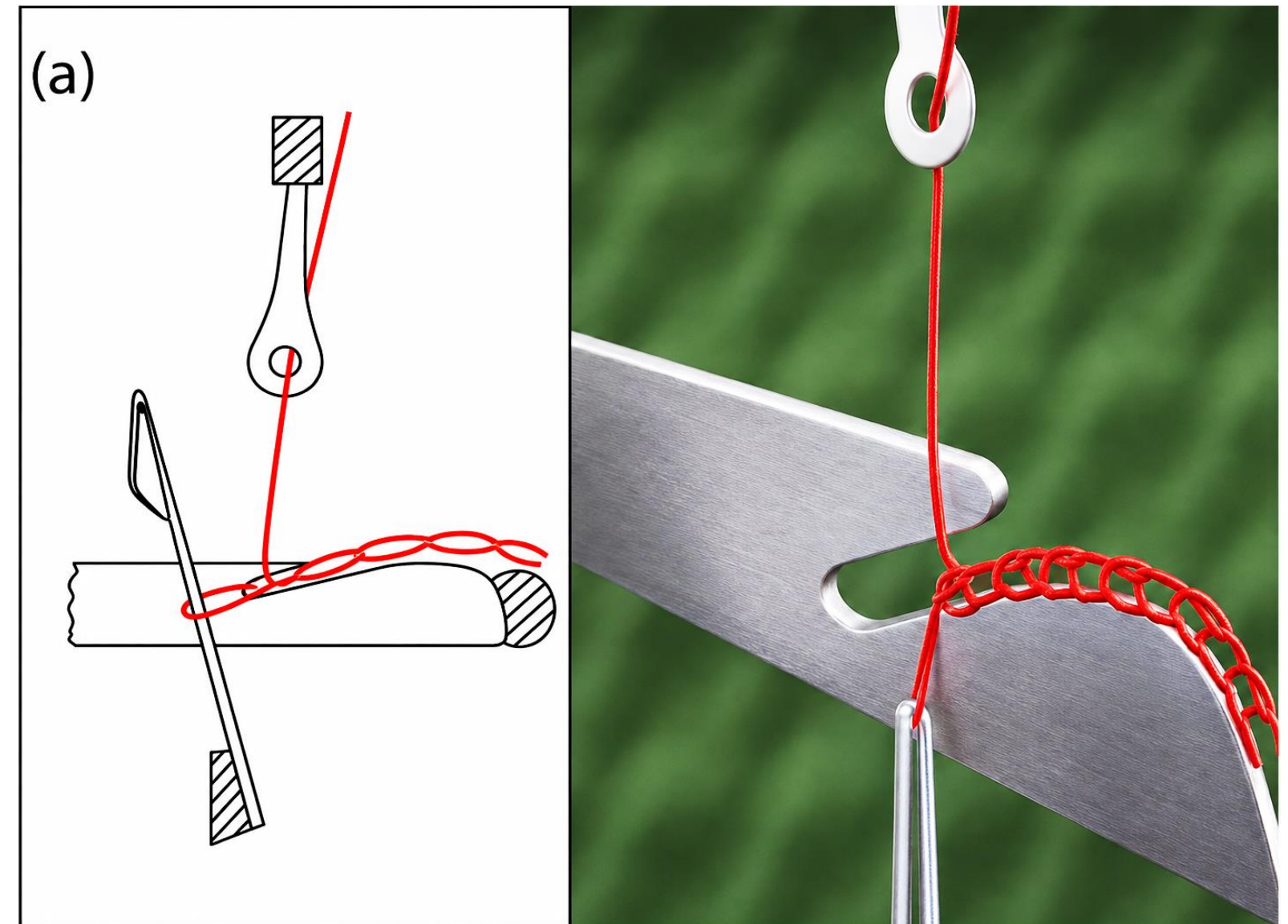
8. Let-off mechanism: The process of releasing the warp yarns according to the requirement of the m/c and speed of cloth roller is the main function of let-off mechanism.
9. Take up mechanism: It helps the produced fabric to wound on the cloth roller in proper tension. There are 3 take-up r/r in this m/c. It also gives proper tension to the warp sheet and controls the speed of warp beam.
10. Link: In the warp knitting m/c the link is used to make design in the knitted fabric. The different links used here has different thickness and thus it helps to produce design.
11. Pattern chain: The pattern chain is the chain of links joined with each other. The pattern chain helps the m/c to produce design.
12. Pattern drum: It is a drum, which gives motion to the pattern chain. There are groove on it and the pattern chain is placed on it. It gets motion from the m/c driving motor through gearing.
13. Comb: In this warp knitting m/c the comb is used to separate the warp yarns coming from the warp beam. It works as the reeds of the weaving m/c and also controls the fabric width.

FUNCTION OF VARIOUS PARTS OF TRICOT WARP KNITTING MACHINE

14. Warp beam: In tricot warp knitting m/c warp beam is used to supply the warp yarns parallelly. There are 8-warp beam in this m/c. 4 beam are on the upper side and 4 are to the downside. These beams are not so big as the weaver's beam.
15. Machine A/C: The m/c a/c is one kind of cooling device, which keeps the m/c parts and the motor from over heating and damage while running continuously.
16. Main shaft: The main shaft of this m/c gives the m/c motion from the motor.
17. Lubrication: The m/c has automatic lubrication system.
18. Controlling unit: The controlling unit is used to control the m/c easily. It has a digital control unit.
19. Toothed belt: In this m/c the toothed belts are used to transfer motion without any slippage.

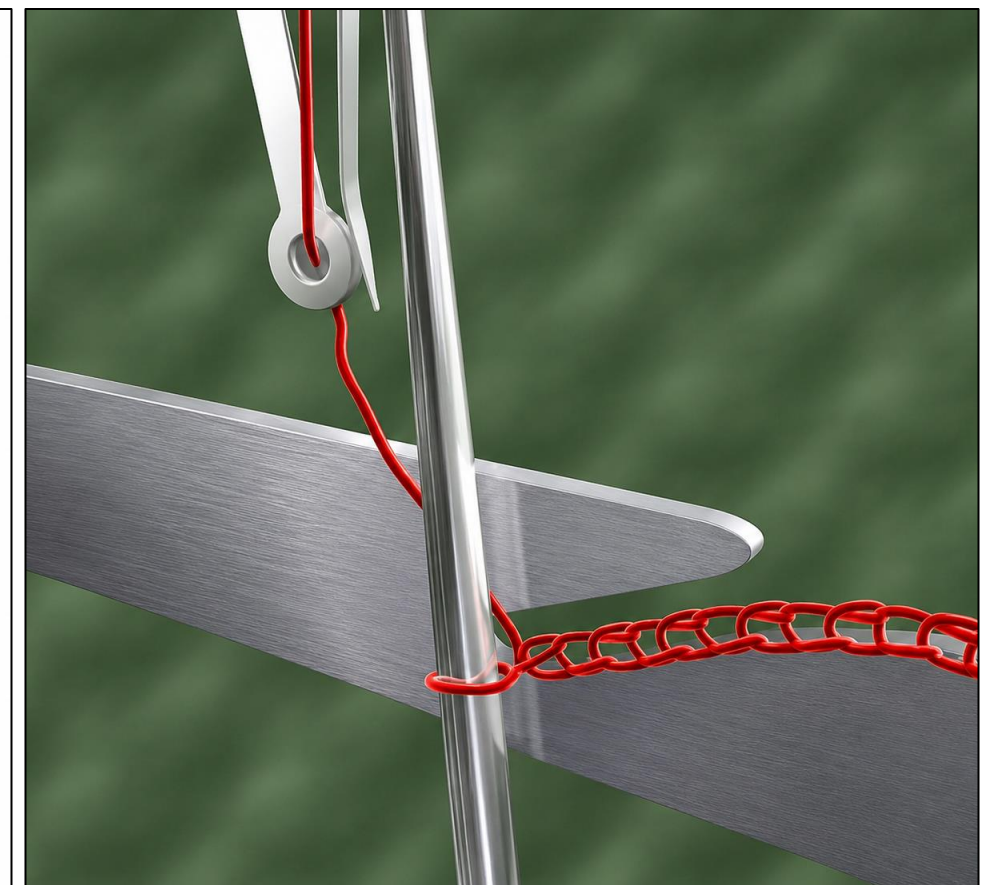
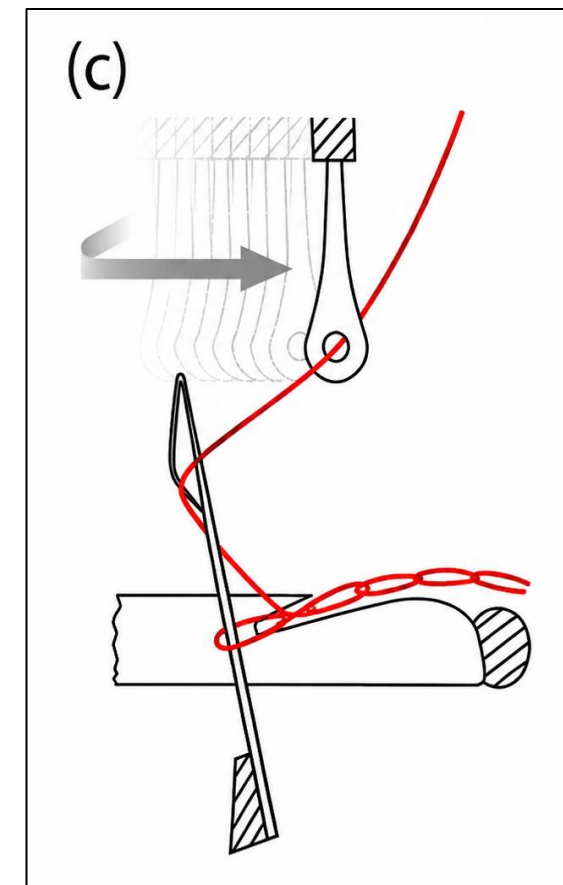
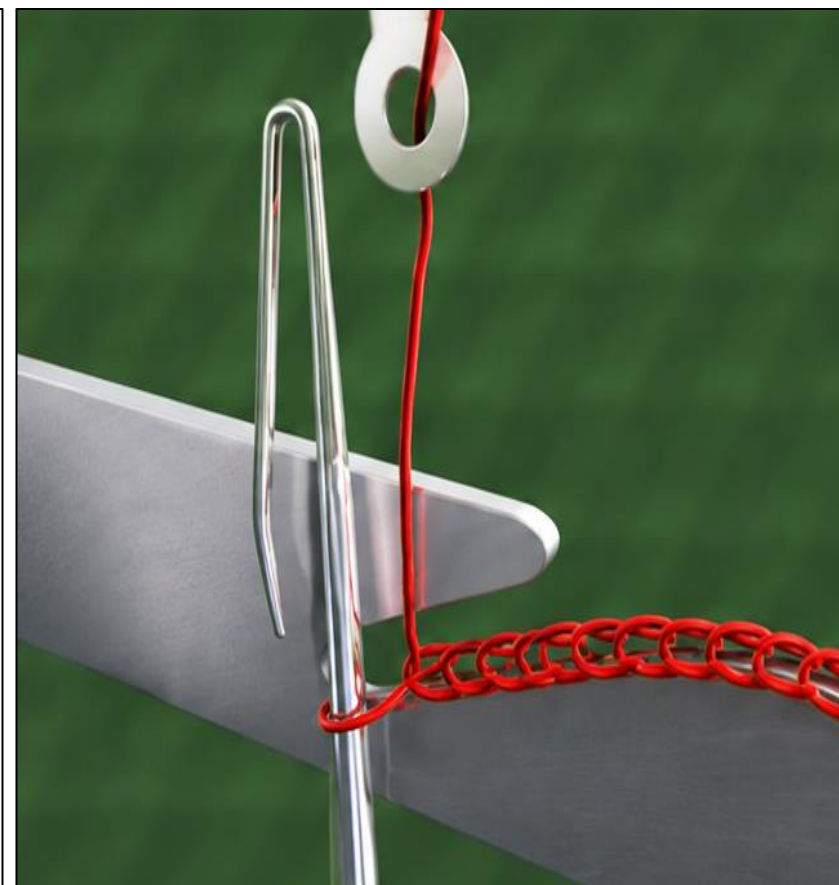
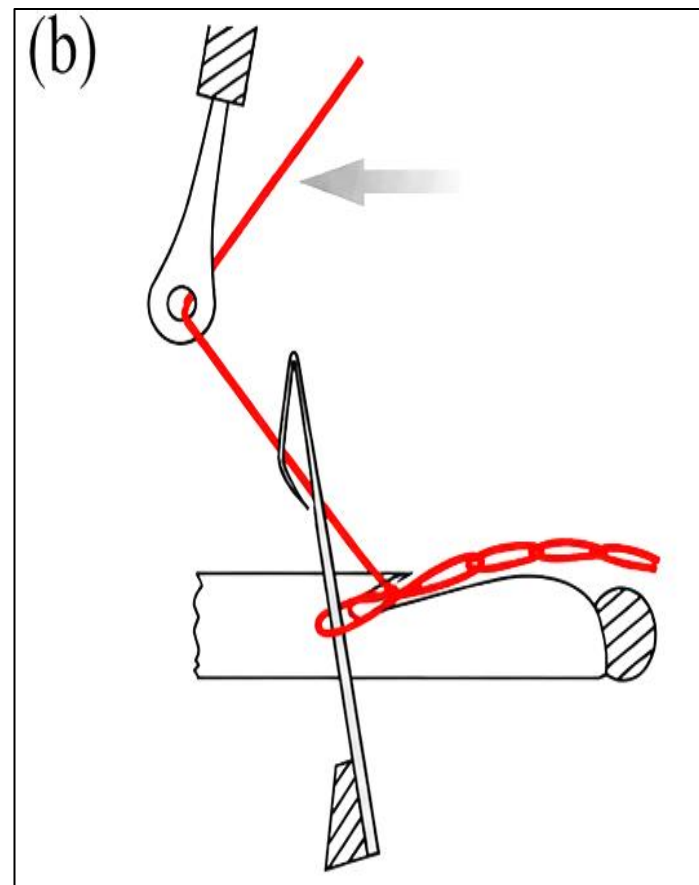
KNITTING ACTION OF BEARDED NEEDLE IN TRICOT WARP KNITTING MACHINE

1. The Rest position (a): The needles have risen to $\frac{2}{3}$ of their full height from knock-over and have their beards towards the back of the machine. The presser is withdrawn and the guides are at the front of the machine with the sinkers forward, holding the old overlaps in their throats so that they are maintained at the correct height on the needle stems.



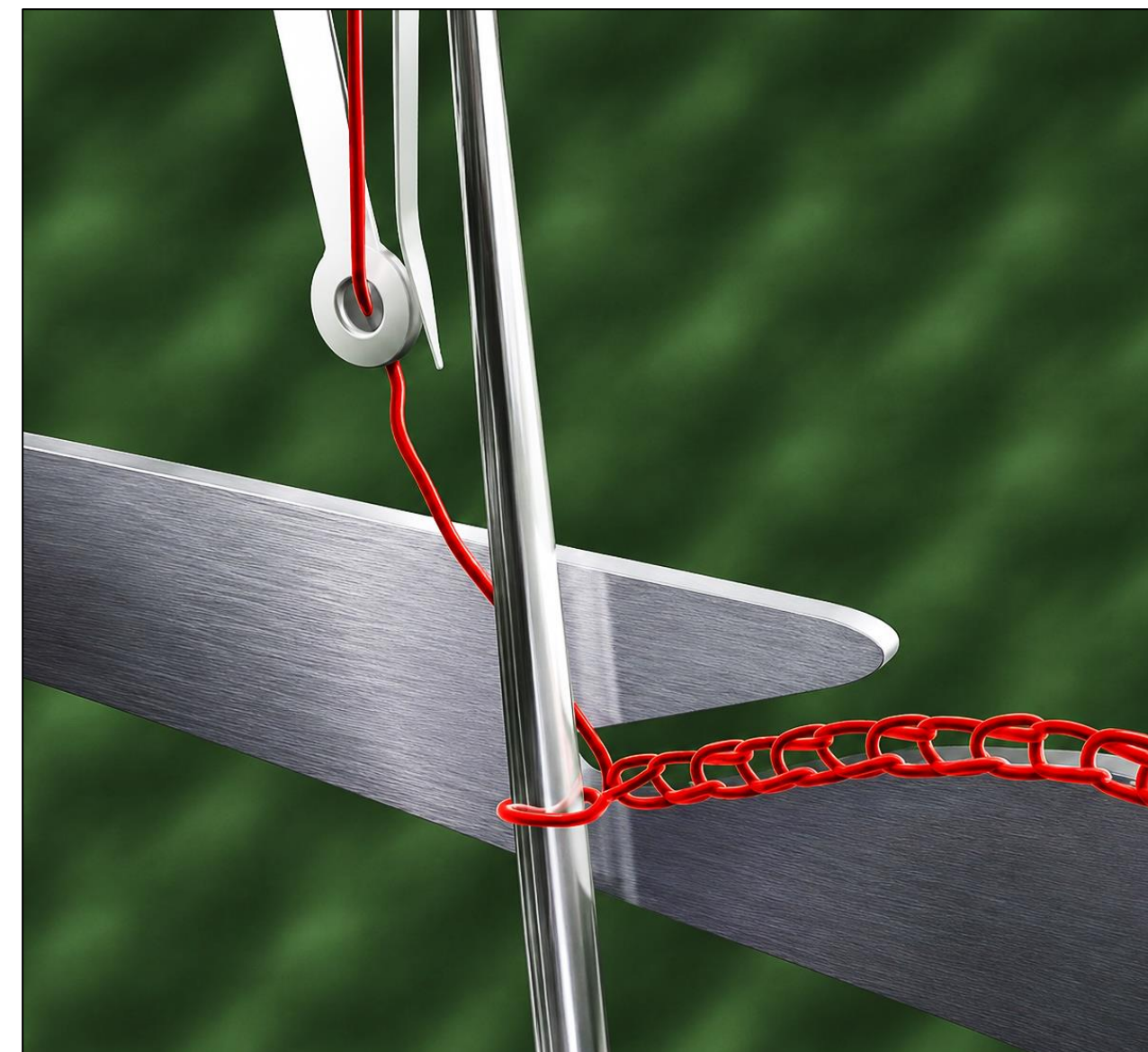
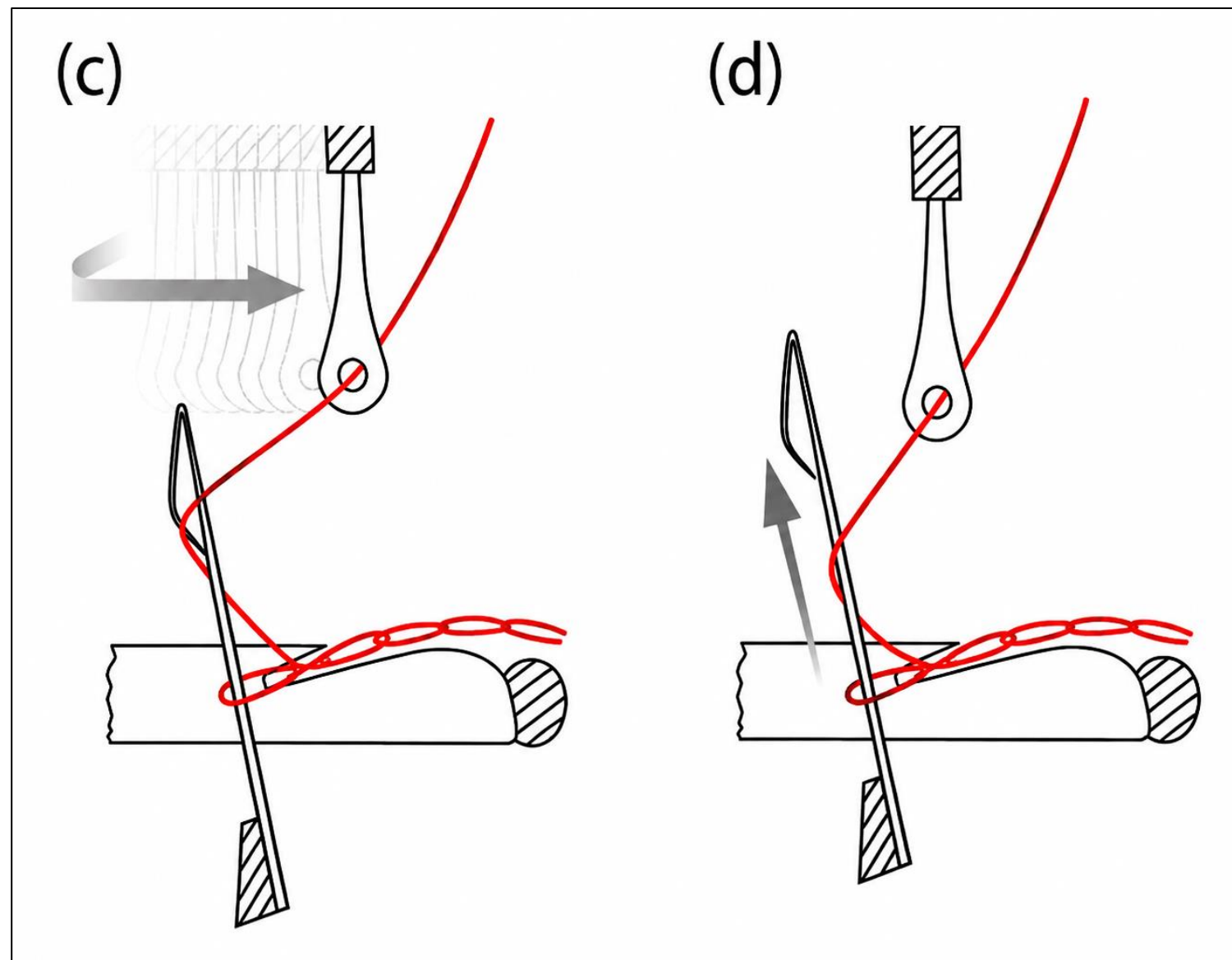
KNITTING ACTION OF BEARDED NEEDLE IN TRICOT WARP KNITTING MACHINE

2. Backward Swing and Overlap Shog (b, c): After swinging through the needles to the beard side, the guides are overlapped across the beards, usually by one needle space in opposite directions.



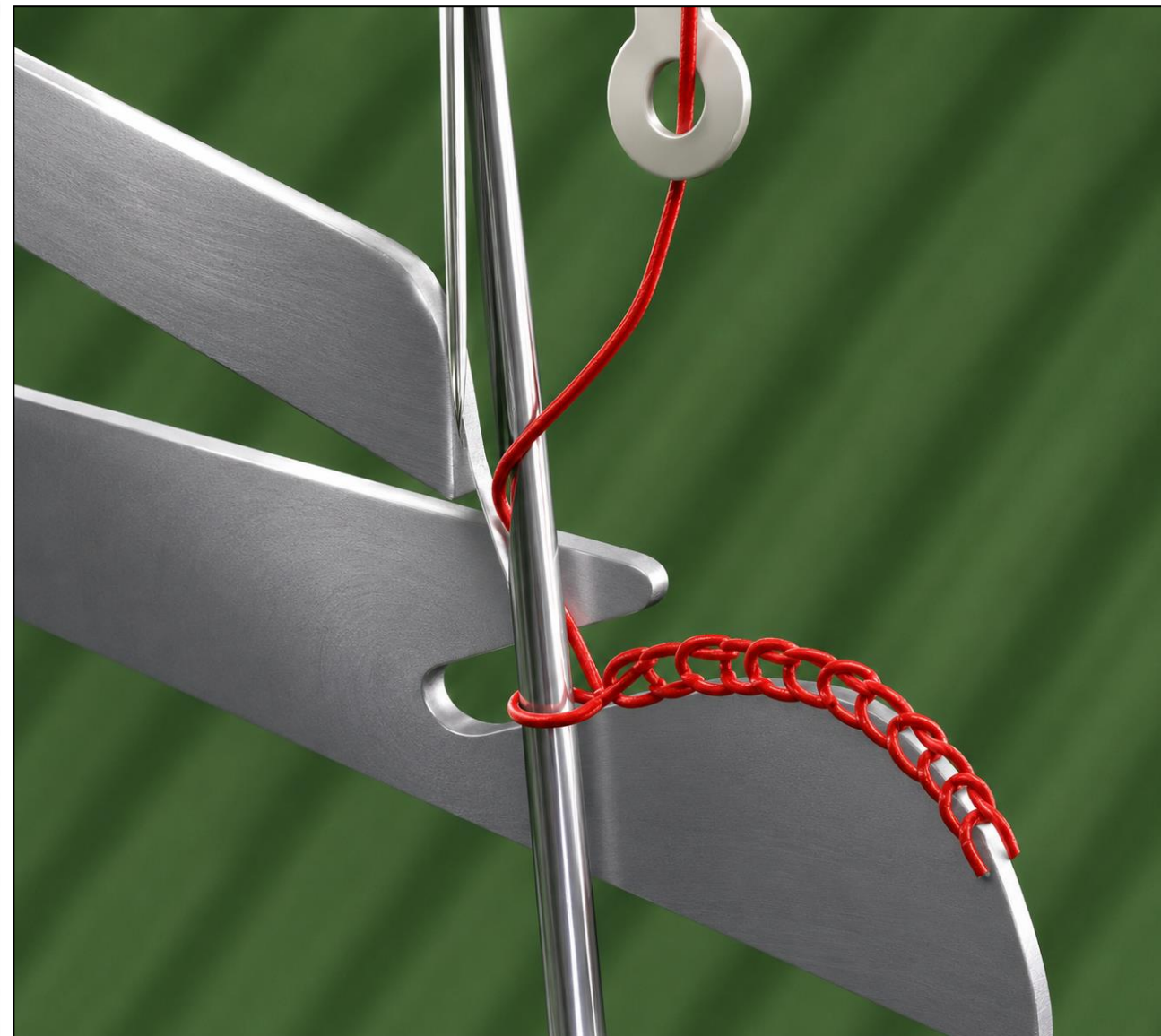
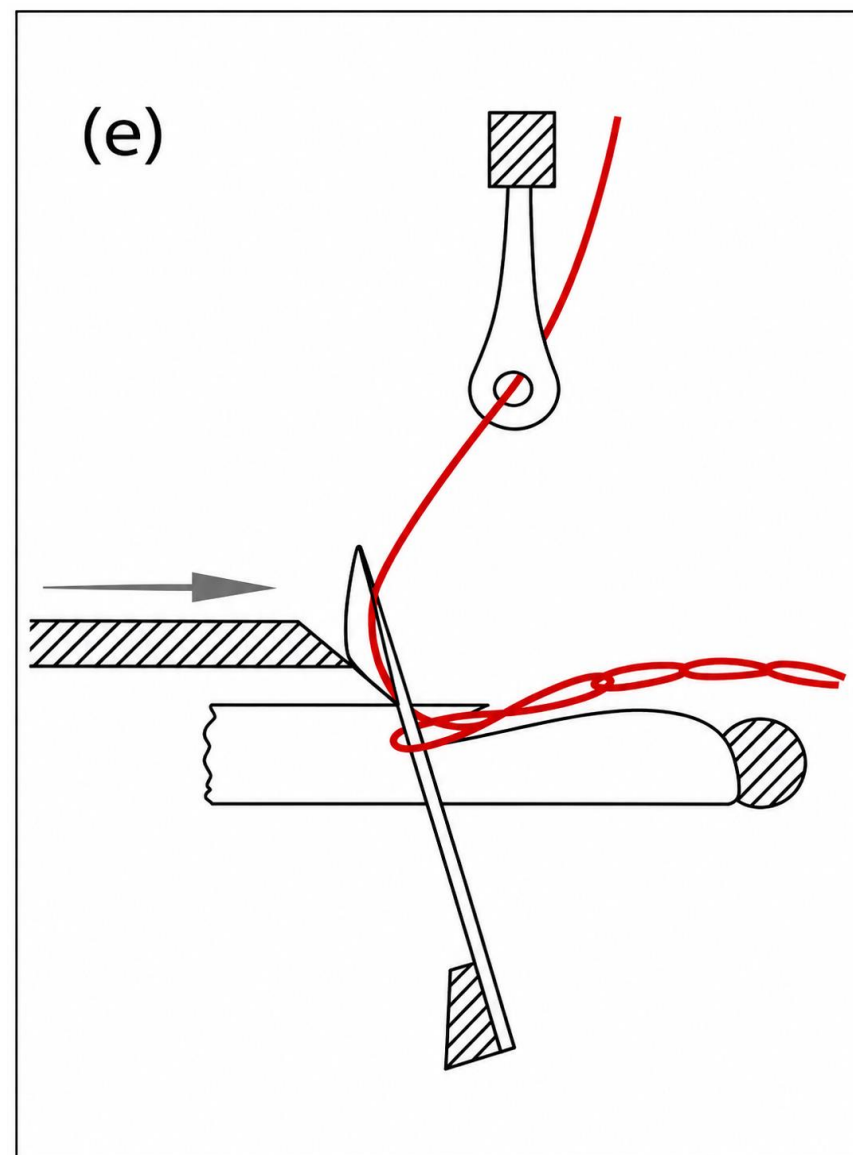
KNITTING ACTION OF BEARDED NEEDLE IN TRICOT WARP KNITTING MACHINE

3. The return swing and second rise (c, d): As the guides swing to the front, the needles rise to their full height so that the newly-formed overlaps slip off the beards onto the stems above the old overlaps.



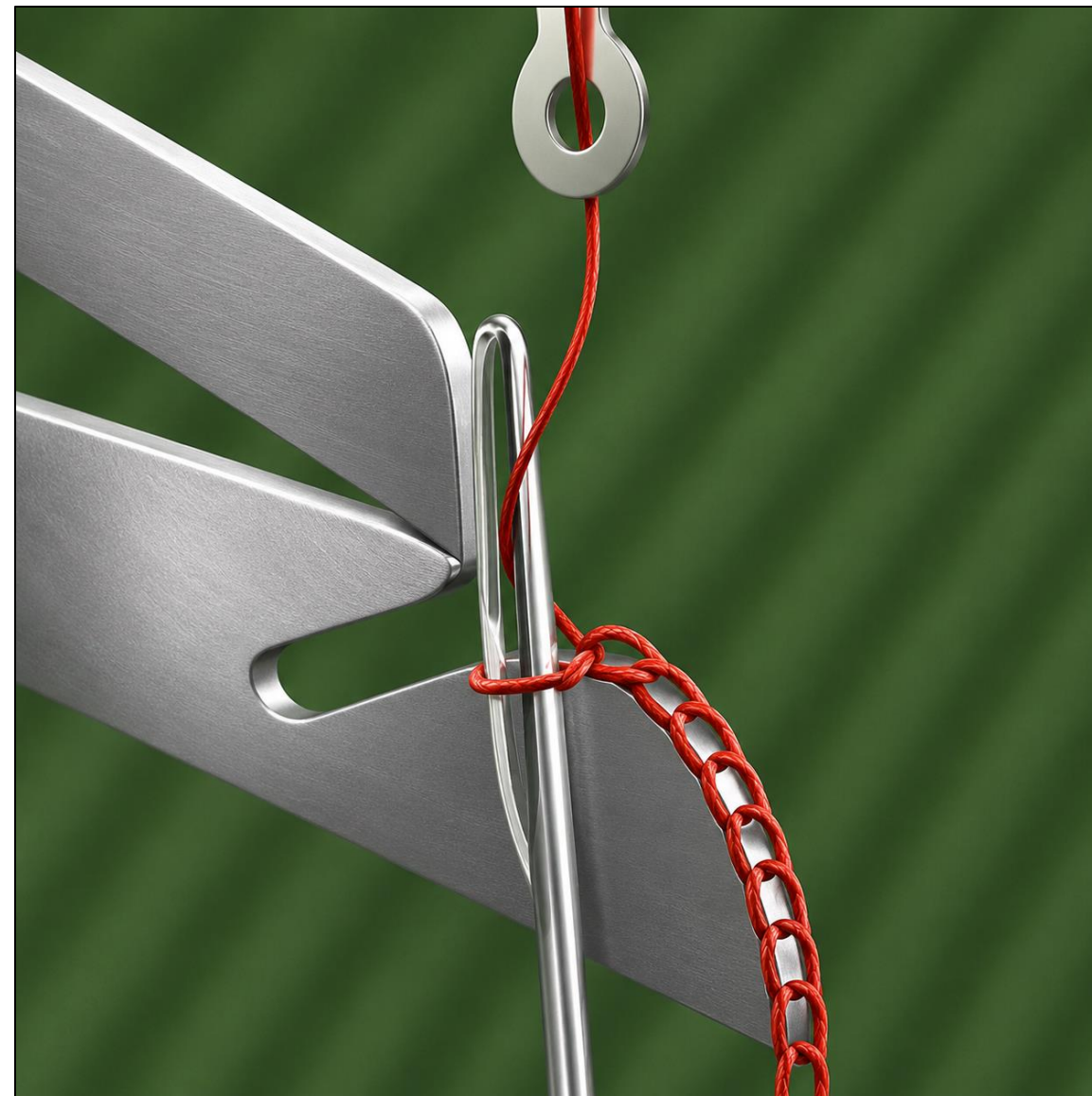
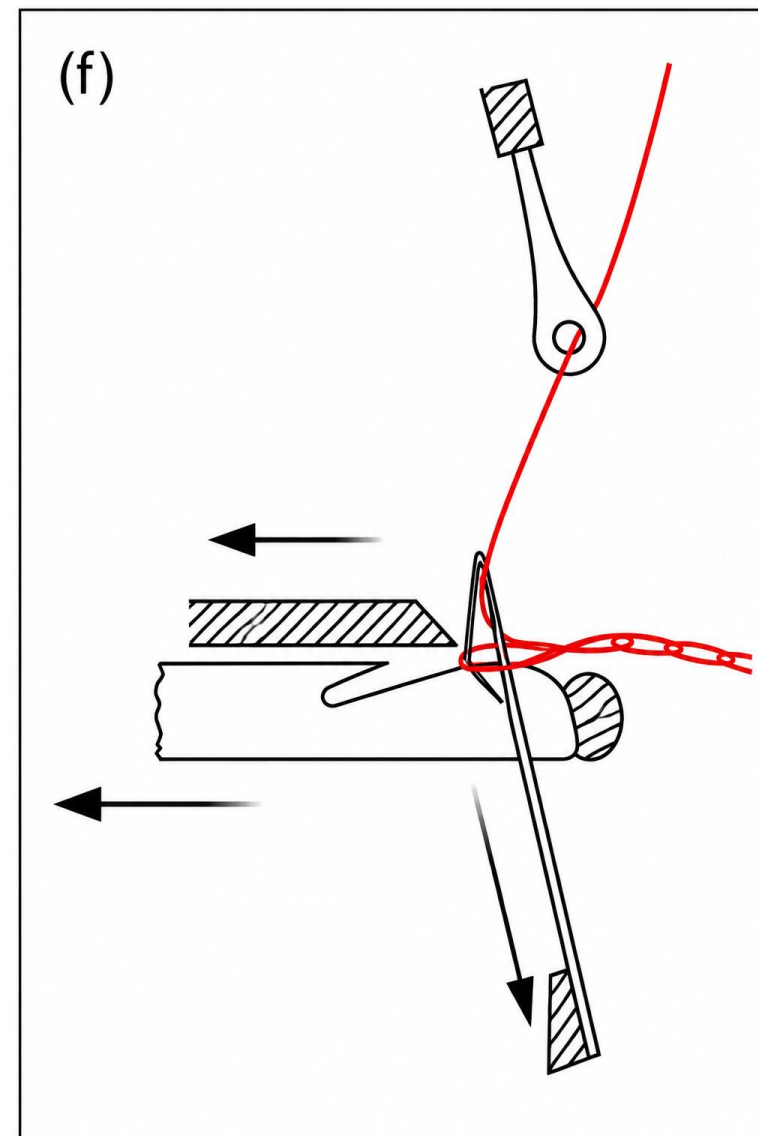
KNITTING ACTION OF BEARDED NEEDLE IN TRICOT WARP KNITTING MACHINE

4. Pressing (e): The needle bar descends so that the open beards cover the new overlaps. There is a slight pause whilst the presser advances and closes the beards.



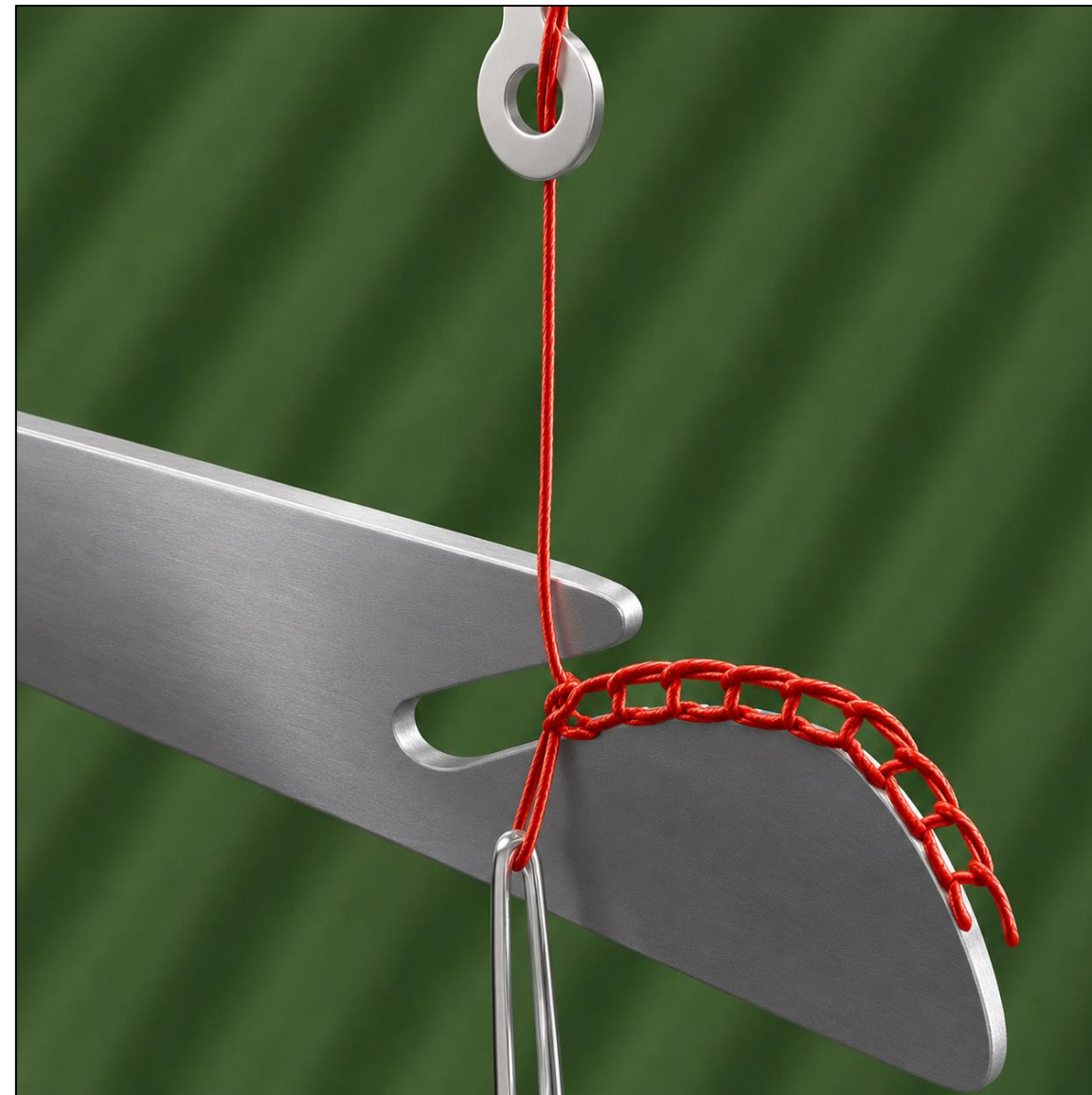
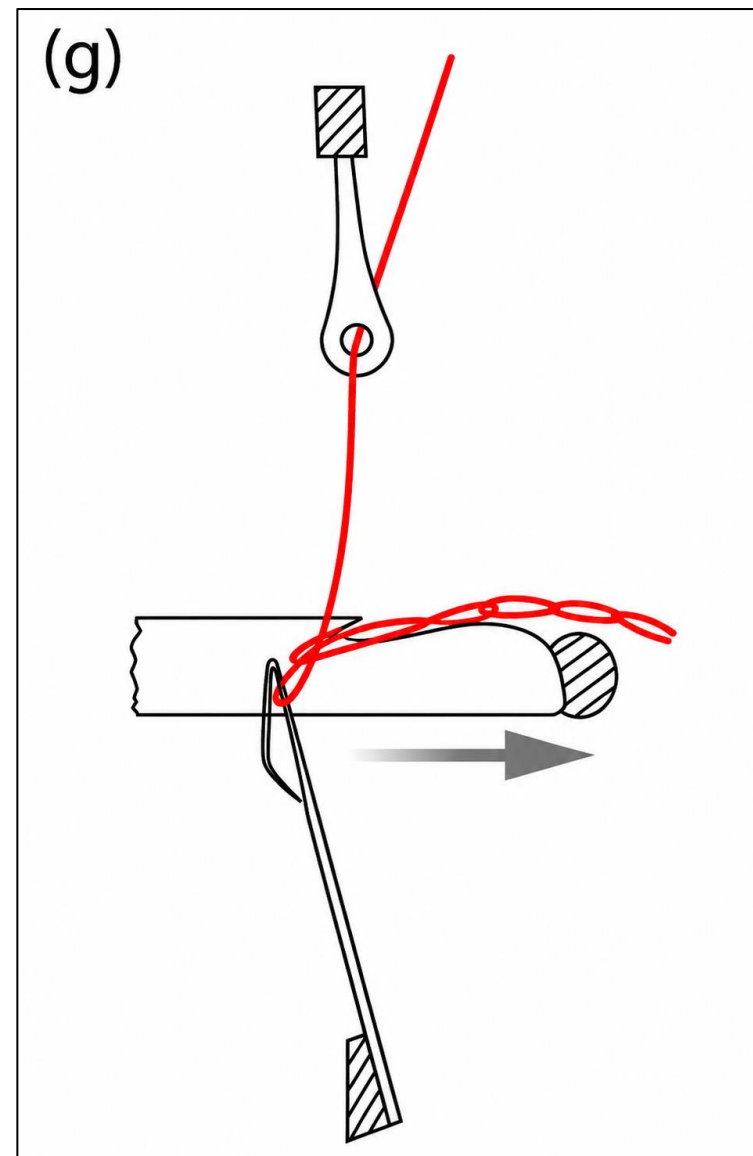
KNITTING ACTION OF BEARDED NEEDLE IN TRICOT WARP KNITTING MACHINE

5. Landing (f): As the **sinkers withdraw**, the upward curve of their bellies lands the **old overlaps** onto the closed beards.



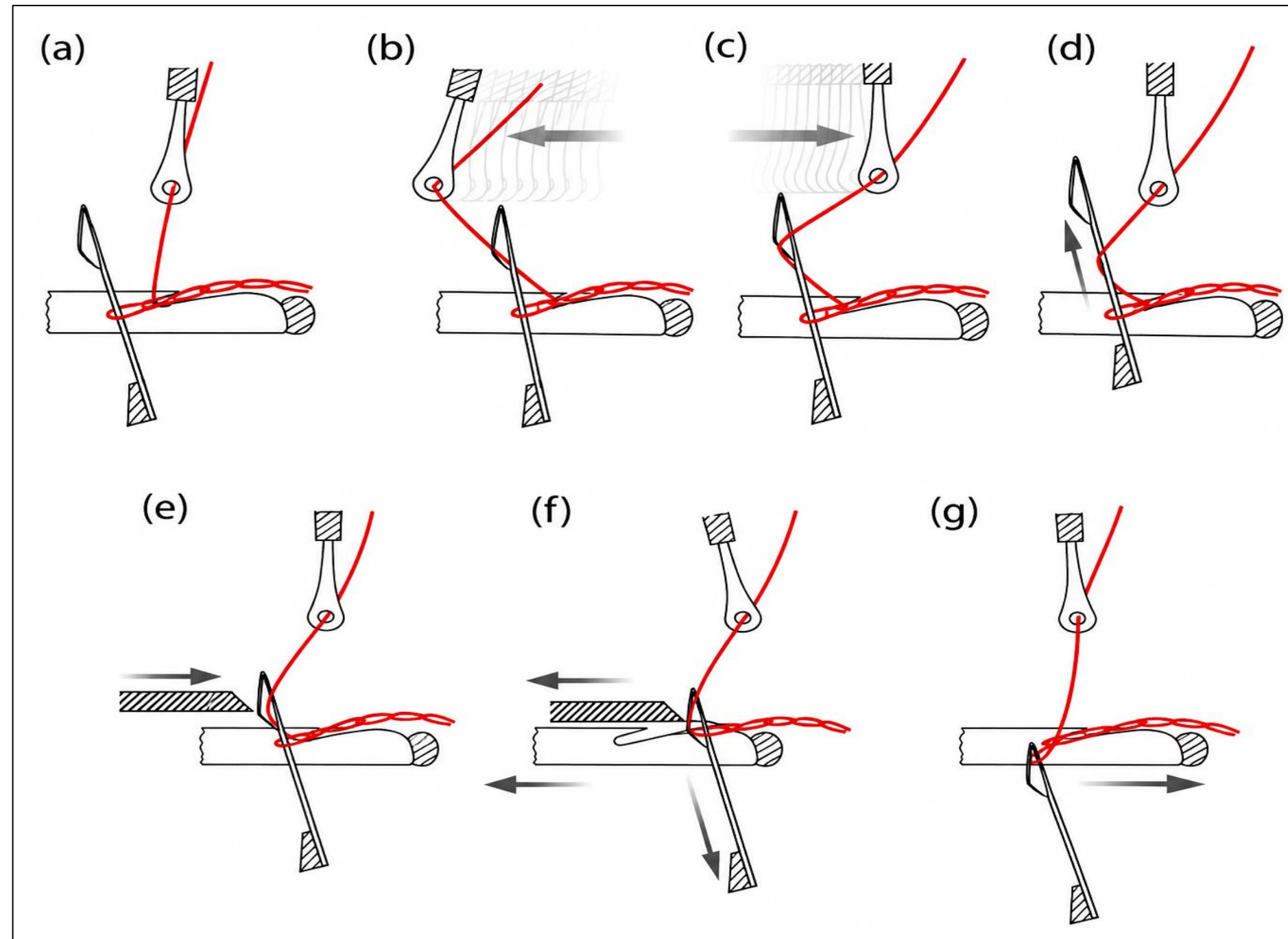
KNITTING ACTION OF BEARDED NEEDLE IN TRICOT WARP KNITTING MACHINE

6. Knock-over and Underlap Shog (g): The **presser is withdrawn** and the **continued descent of the needle bar** causes the **old overlaps to be knocked-over** as the heads of the needles descend below the upper surface of the sinker bellies. The underlap shog which can occur at any time between pressing and knock-over usually occurs in opposite directions on the two guide bars.



KNITTING ACTION OF BEARDED NEEDLE IN TRICOT WARP KNITTING MACHINE

The sinkers now move forward to hold down the fabric loops and push them away from the ascending needles, which are rising to the rest position.



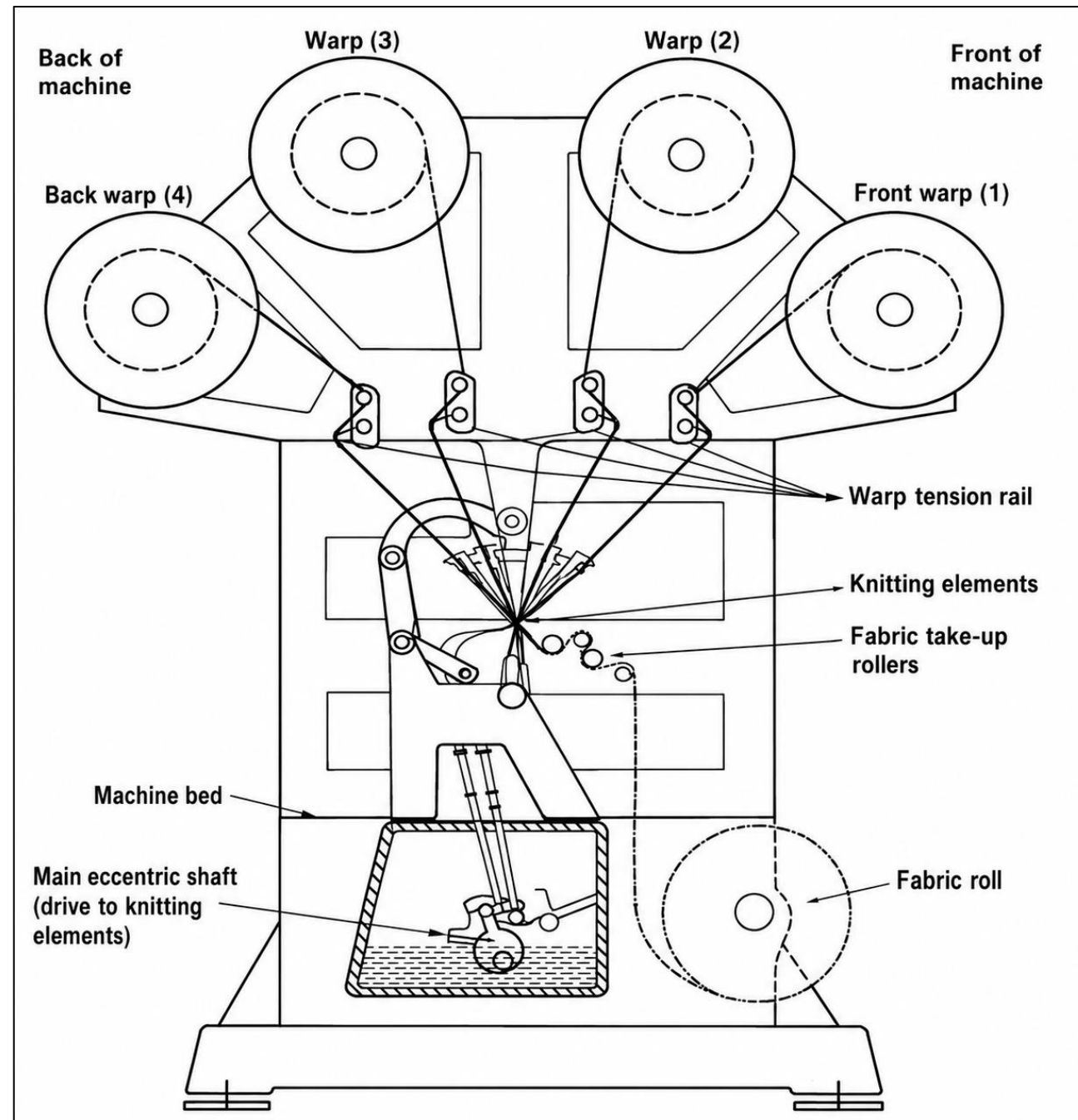
FEATURES OF RASCHEL WARP KNITTING MACHINE

1. Raschel machine used latch needles together with a wire or blade. Presently compound needle are also used at Raschel machine.
2. It has a gauge expressed in Needles per 2 Inches (5cm). For example, a 36E Rachel m/c will have 18 NPI. There is a wide gauge ranges from E1 to E32.
3. Their chain links are usually numbered in even numbers 0, 2, 4,etc. generally with 2 links/course.
4. Rachel sinkers only performs the function of holding down the loops whilst the needles rise
5. Rachel sinkers are not joined together by a lead across their ends nearest to the needle bars so they can move away clear of the needles towards the back of the m/c for the rest of the knitting cycle.
6. Their needle trick verge acts as a fabric support ledge and knock over surface.
7. The fabric is drawn downwards from the needles, almost parallel to the needle bar at an angle of 120° - 160° by a series of take-down rollers. This creates a high take up tension, particularly suitable for open fabric structure such as laces and nets.
8. The warp beams are arranged above the needle bar, centered over the rocker shaft, so that the warp sheets pass down to the guide bars on either side of it.

FEATURES OF RASCHEL WARP KNITTING MACHINE

9. The beams are placed above the machine so that it is accessible at the front for the fabric inspection.
10. Mechanical attention are carried out at the back of the machine.
11. The guide bars are threaded, connecting with the middle bars and working outwards from the either side of the rocker shaft.
12. The guide bars are numbered from the front of the m/c.
13. There is accommodation for at least four 32-inch diameter beams or large numbers of small diameter pattern bars.

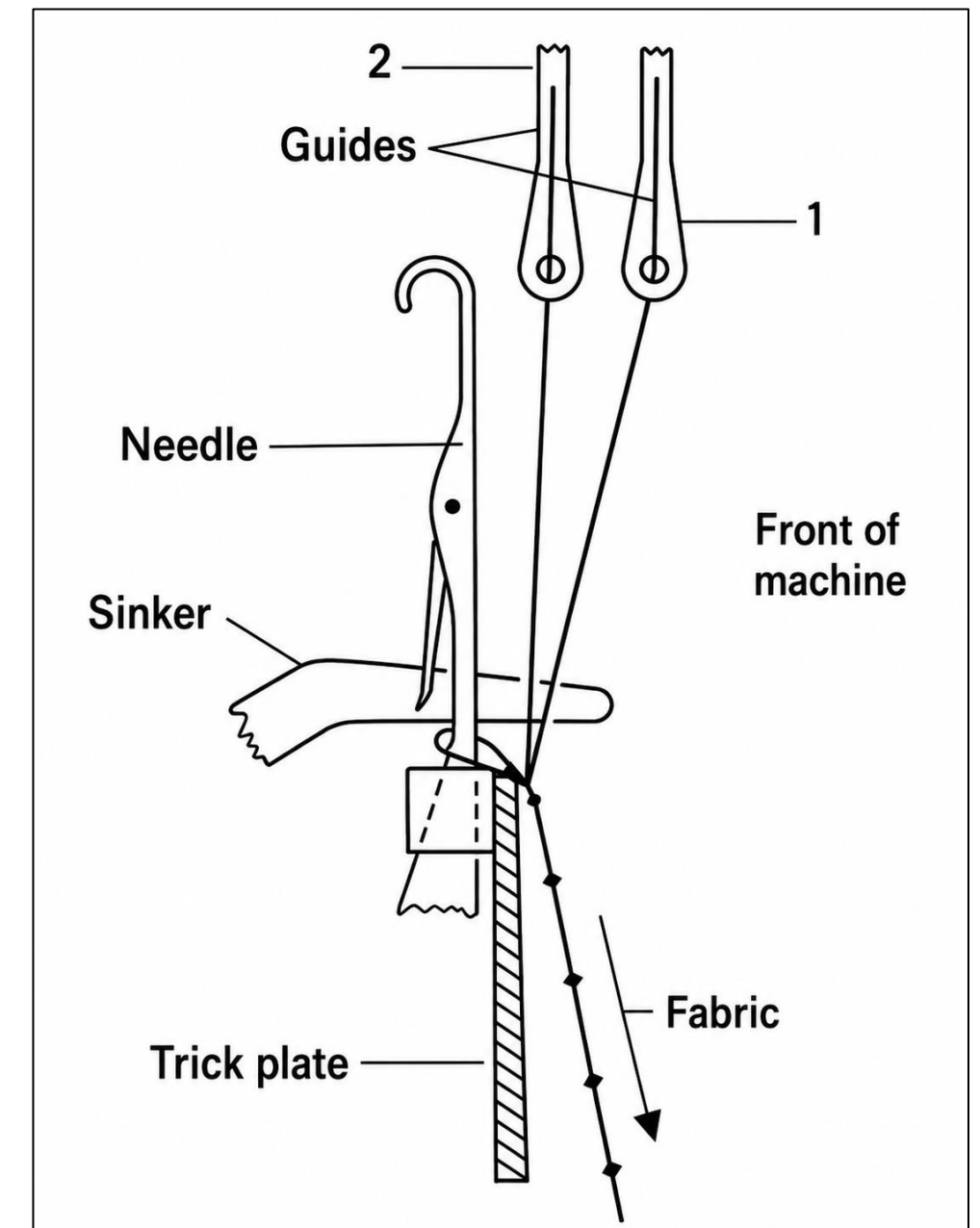
FEATURES OF RASCHEL WARP KNITTING MACHINE



KNITTING ACTION OF LATCH NEEDLE IN RASCHEL WARP KNITTING MACHINE

Raschel needles tend to have longer latches than weft knitting machine needles, to ensure that the wrapped yarns of the overlap goes onto and not below the open latch. There is a **trick-plate** extending the full width of the machine, whose walls preserve the needle spacing and whose verge provides an edge for a clean knock-over. Holding-down sinkers that are thin blades, unleaded at their forward edges, move in a horizontal plain over the top of the trick-plate.

1. Holding down: The guide bars are at the front of the machine, completing their underlap shog. The sinker bar moves forward to hold the fabric down whilst the needle bar starts to rise from knock-over.
2. Clearing: As the needle bar rises to its full height, the old overlaps slip down onto the stems after opening the latches, which are prevented from flicking closed by latch wires. The sinker bar then starts to withdraw to allow the guide bars to overlap.

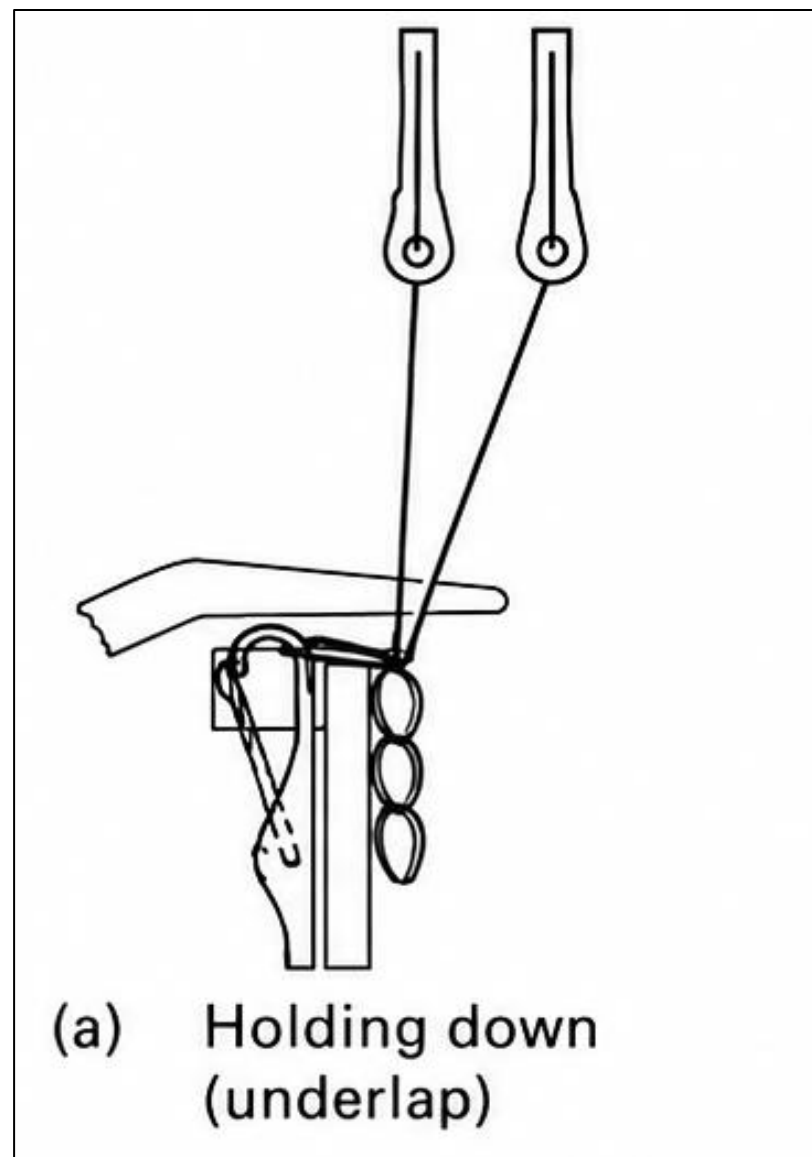


KNITTING ACTION OF LATCH NEEDLE IN RASCHEL WARP KNITTING MACHINE

3. Overlap: The guide bars swing to the back of the machine and then shog for the overlap. Knitting action of single needle bar latch needle Raschel warp knitting machine.
4. Return swing: As the guide bars swing to the front, the warp threads wrap into the needle hooks.
5. Latch closing: The needle bar descends so that the old overlaps contact and close the latches, trapping the new overlaps inside. The sinker bar now starts to move forward.
6. knocking-over and underlap: As the needle bar continues to descend, its head passes below the surface of the trick-plate, drawing the new overlap through the old overlap which is cast-off and as the sinkers advance over the trick-plate, the underlap shog of the guide bar is commenced.

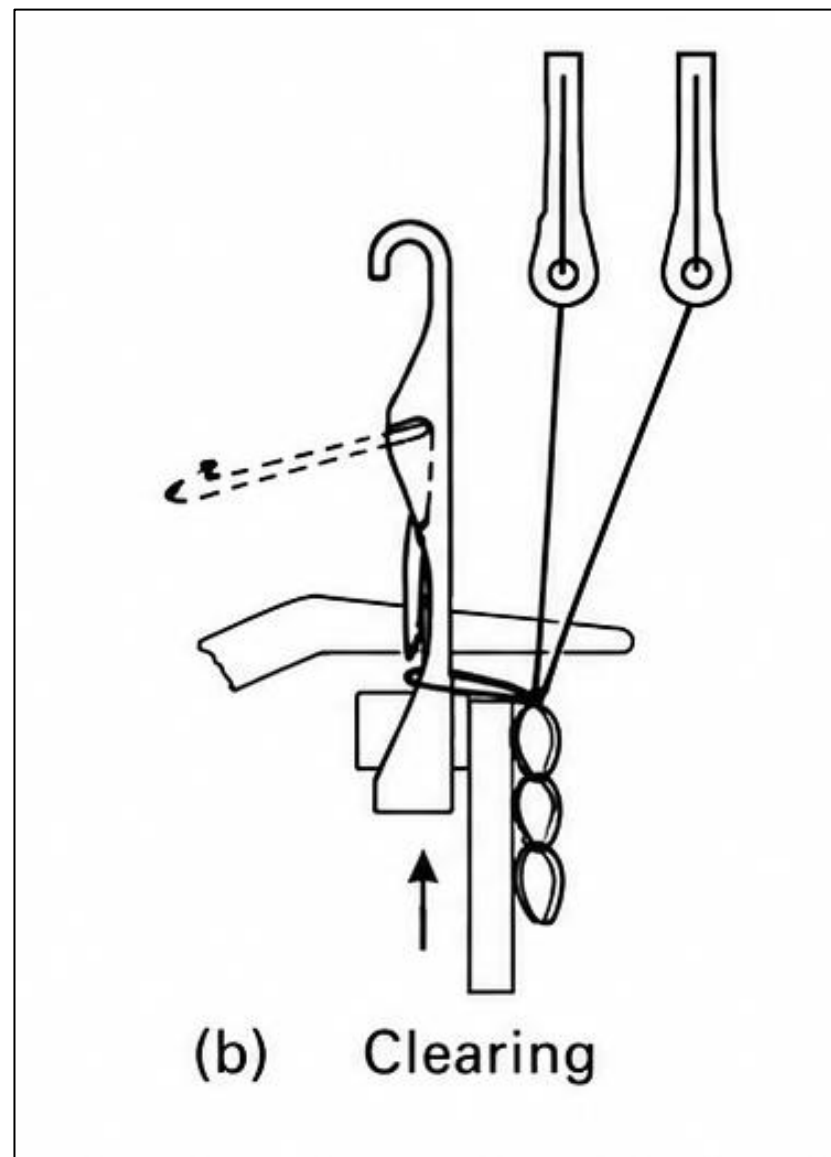
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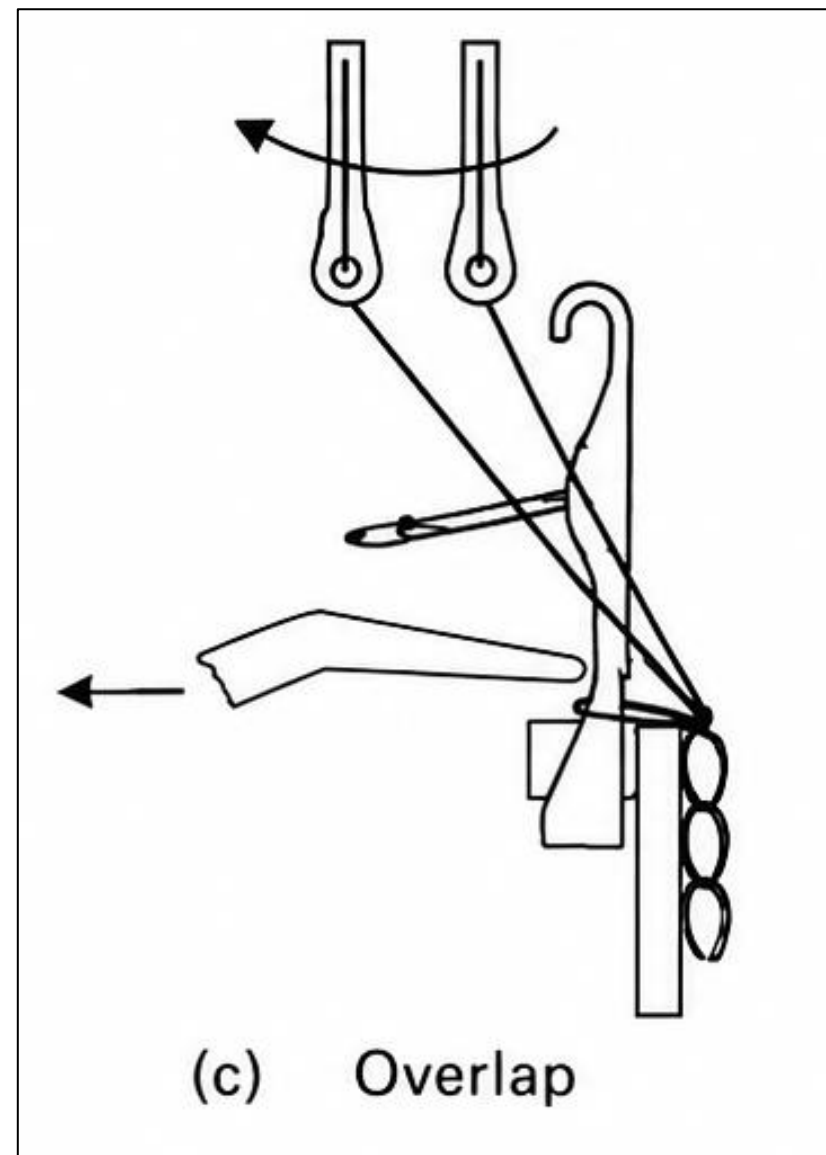
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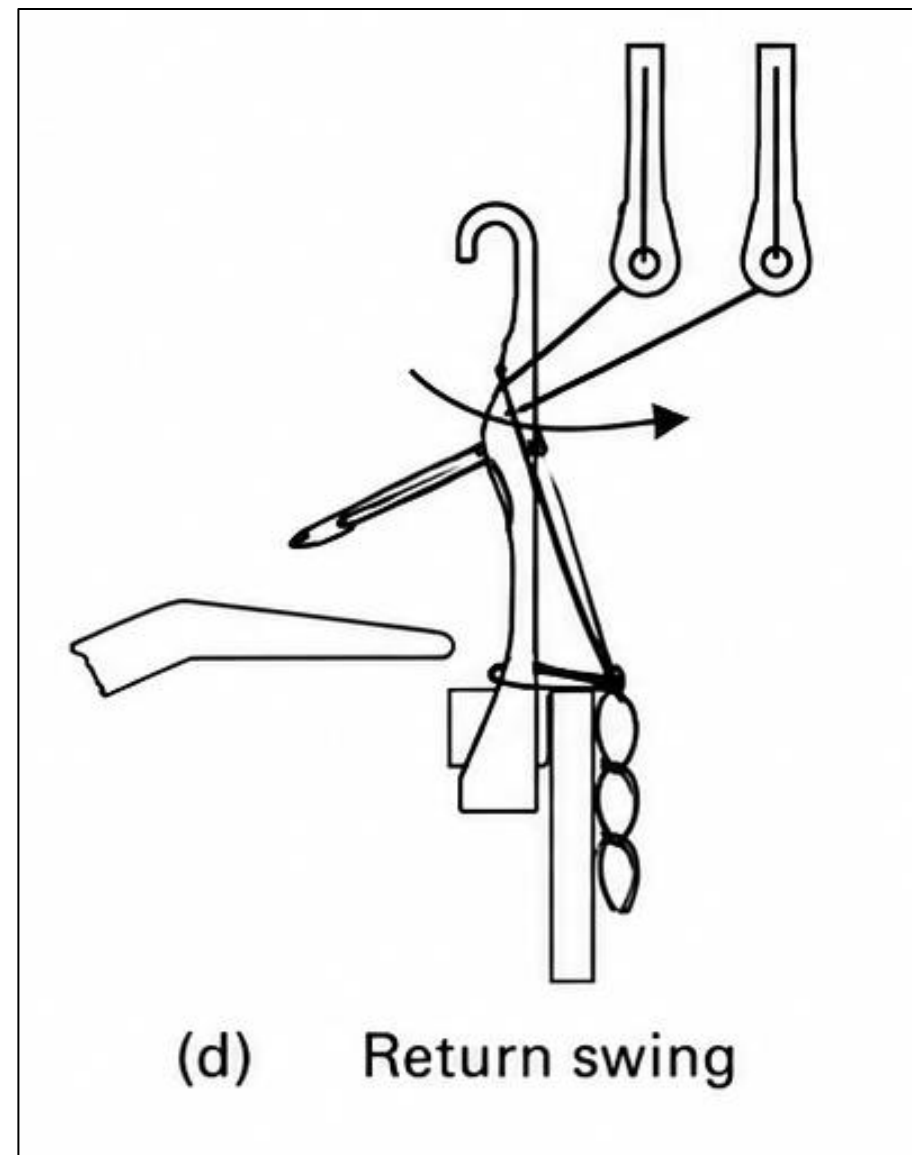
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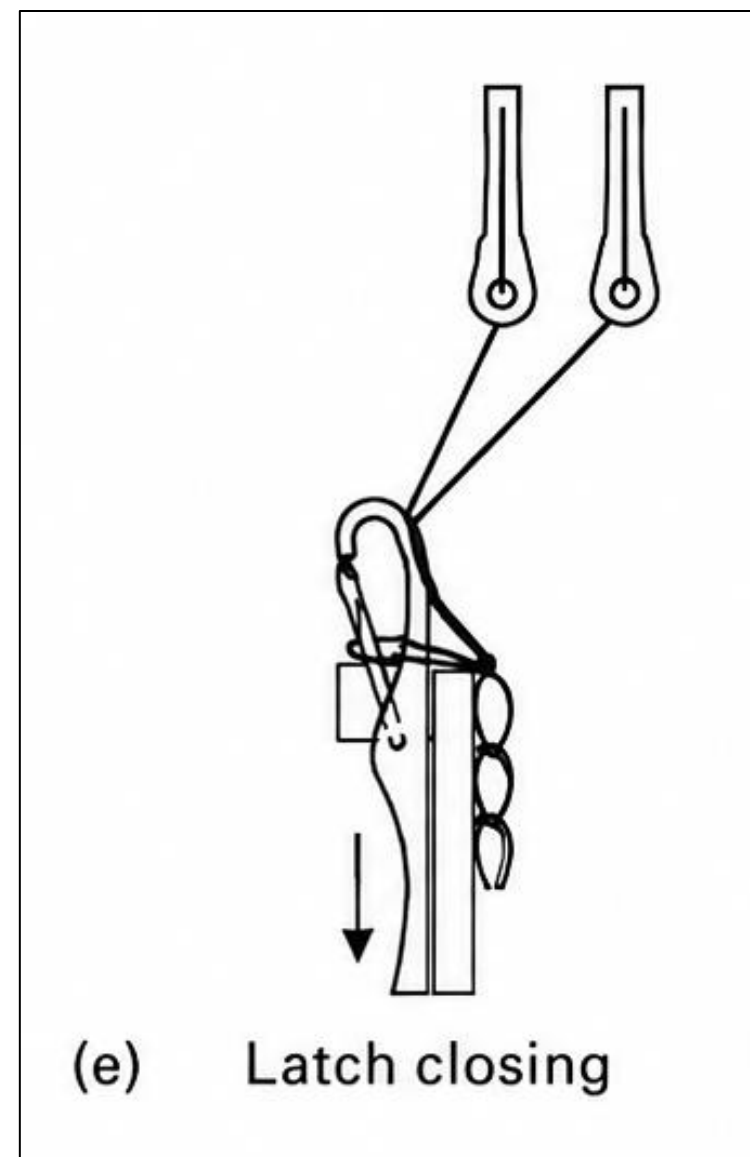
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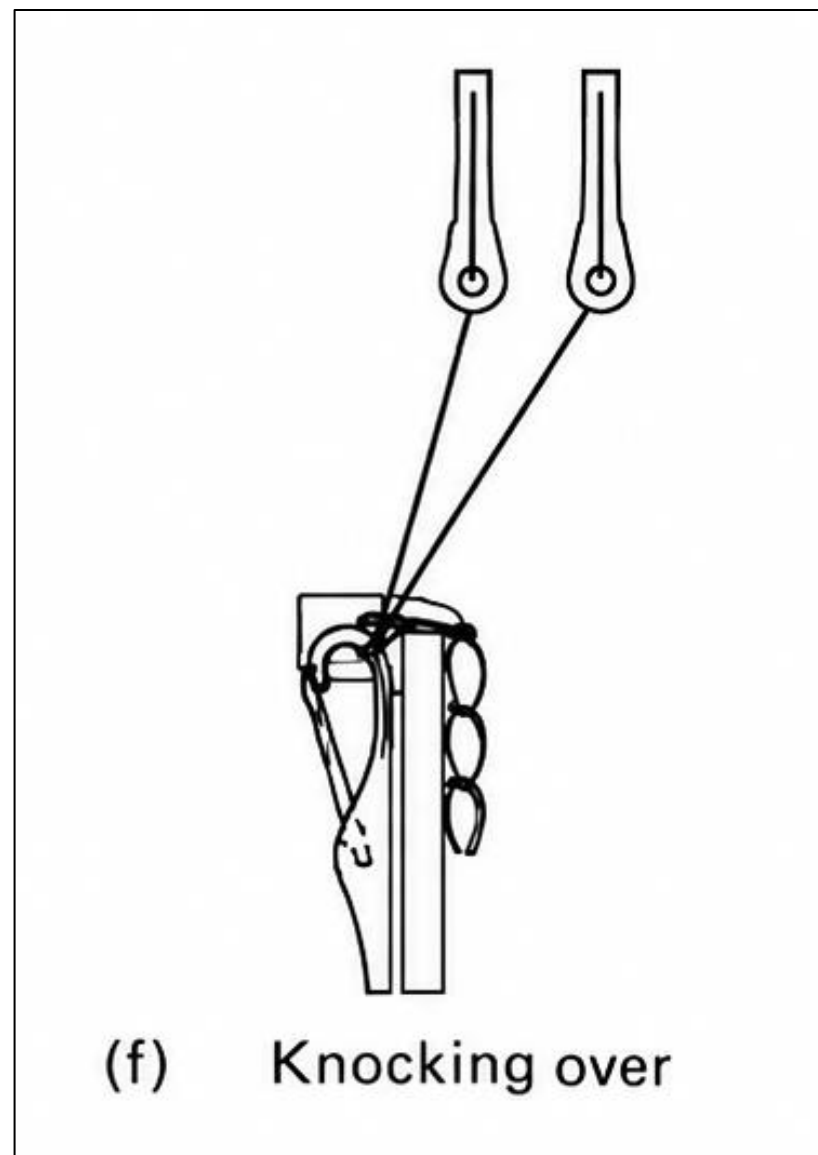
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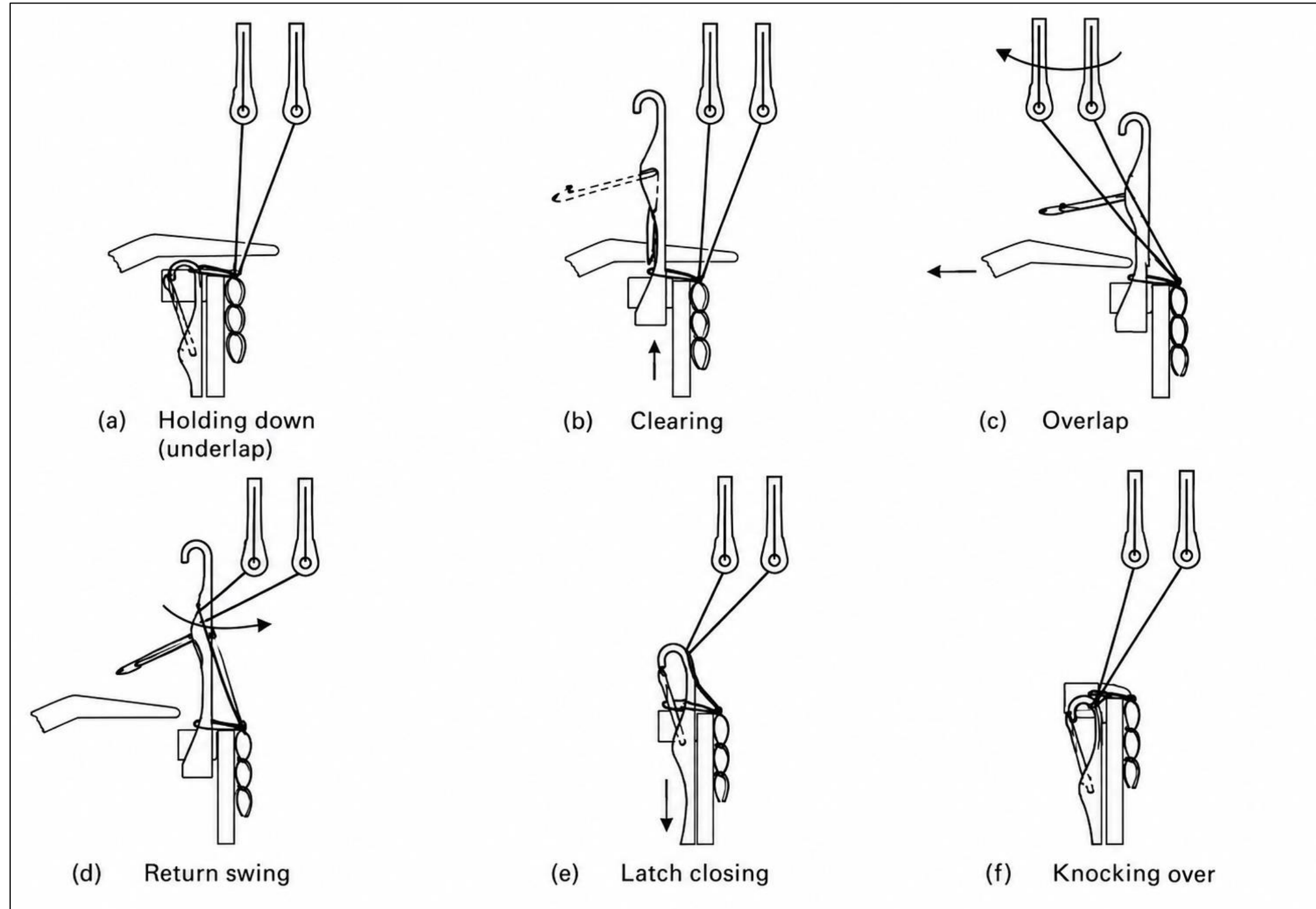


KNITTING ACTION OF LATCH NEEDLE IN RASCHEL WARP KNITTING MACHINE

6. knocking-over and Underlap: As the needle bar continues to descend, its head passes below the surface of the trick-plate, drawing the new overlap through the old overlap which is cast-off and as the sinkers advance over the trick-plate, the underlap shog of the guide bar is commenced.



KNITTING ACTION OF LATCH NEEDLE IN RASCHEL WARP KNITTING MACHINE



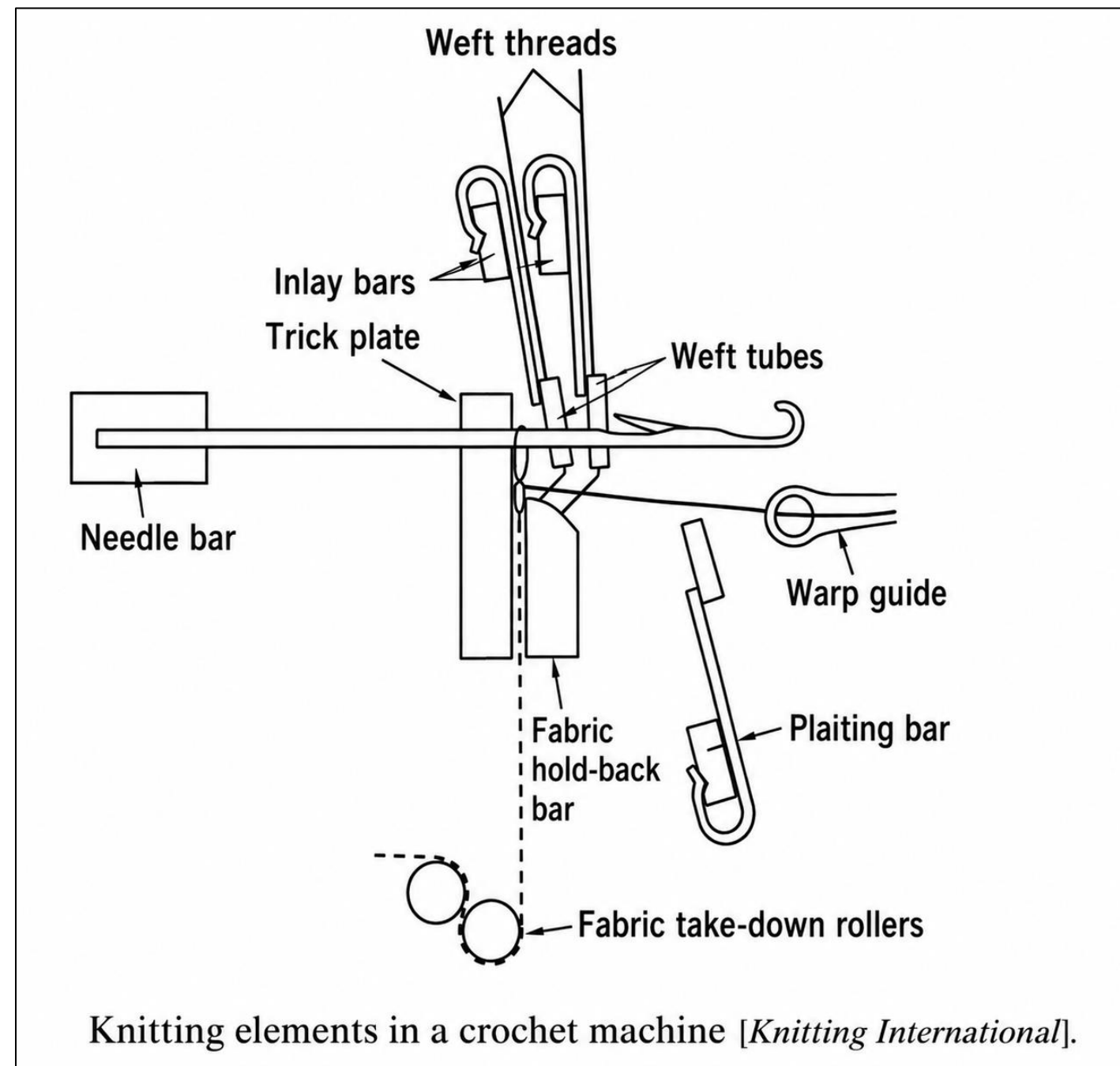
FEATURES OF CROCHET WARP KNITTING MACHINE

1. A single horizontal needle bar whose simple reciprocating action can be used to operate individually tricked latch, carbine or embroidery needles.
2. The patent or carbine bearded needle is used for fine structures and has a side ways crimped beard placed in a permanently pressed position.
3. Although warp threads can only be fed into the beard from the left, the old overlaps are automatically cleared and landed by the movement of the needle. The needles can be arranged for coarser gauges or for fancy set-outs, when the floating inlay threads may be cut to produce separated fringed edgings.
4. No sinkers; instead a fixed hold back bar is fitted in from of the knock over verge to prevent the fabric moving out with the needles.
5. The closed lap pillar stitches and inlay threads controlled and supplied as separate warp and weft respectively.

FEATURES OF CROCHET WARP KNITTING MACHINE

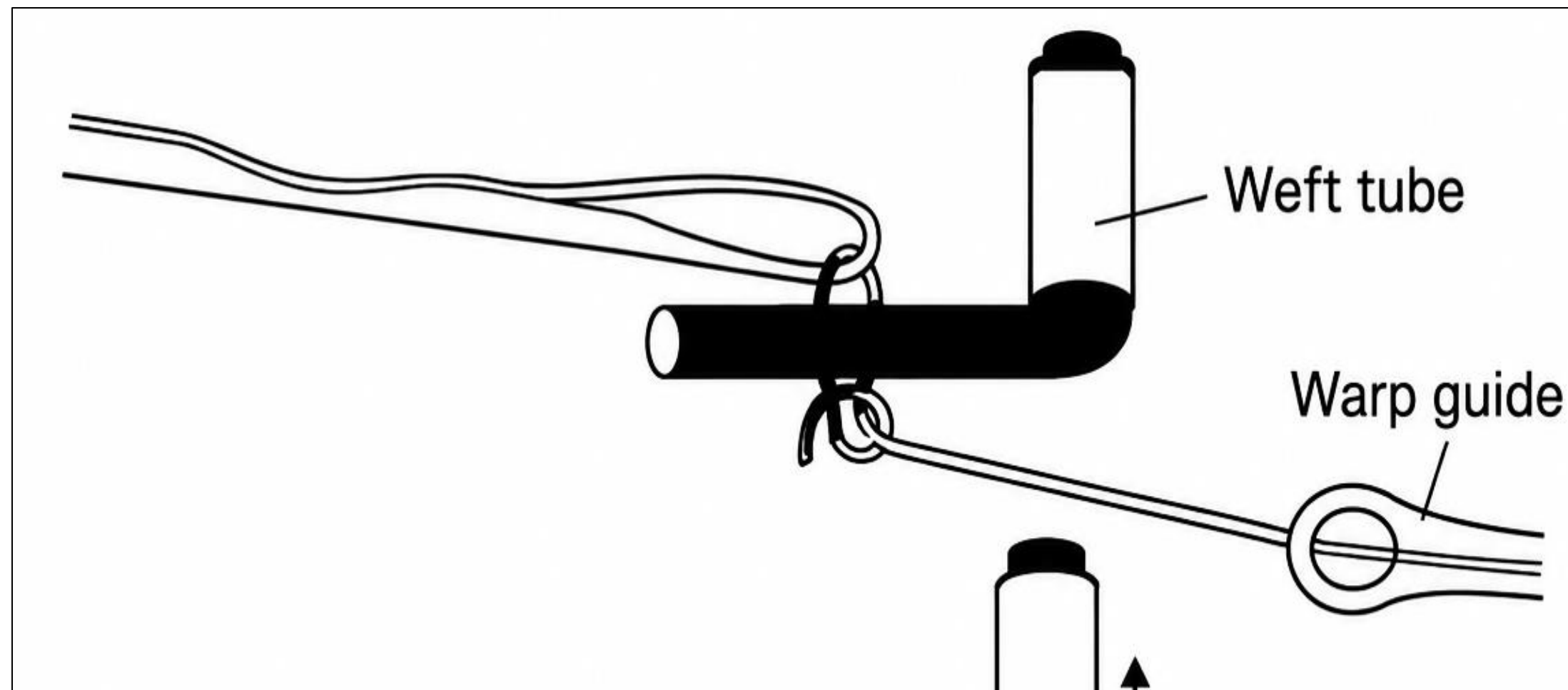
6. The weft yarn, often placed above and towards the back of the machine, supplying the carrier tubes which are clipped to the spring-loaded inlay bars.
7. Special arrangements are available for producing fancy effects such as cut or uncut fringe edges, pile, braiding and snail shell designs.
8. Gauge often expressed in needles per centimeter are between 2 and 10 (E5 to E24). Knitting widths of crochet machine may vary between 16 and 122 inches.
9. Machine runs at speeds between 200 and 350 courses per minute.
10. Crochet machine can process a range of filament yarn from 20dtex to 1000dtex.
11. The warp yarn is often placed low at the front of the machine.

KNITTING ELEMENTS OF CROCHET WARP KNITTING MACHINE



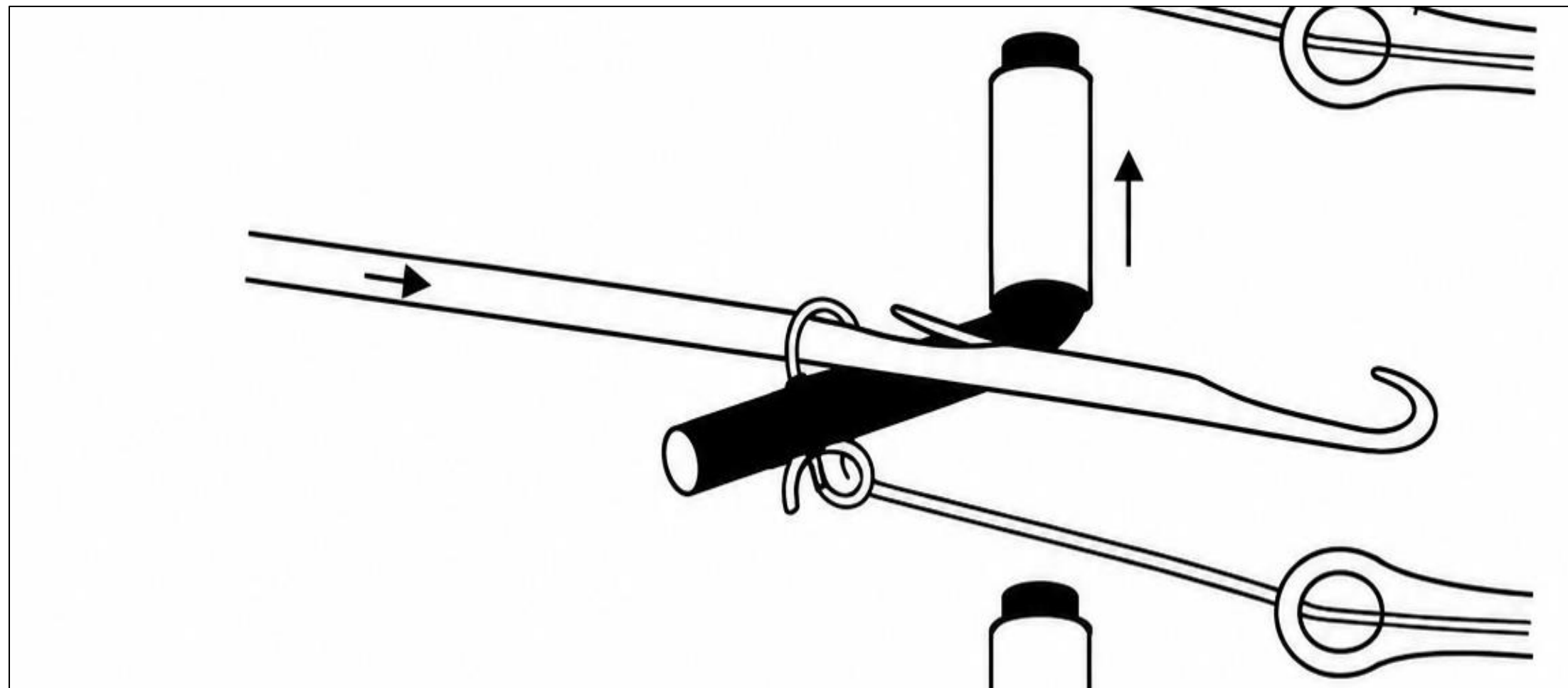
KNITTING ACTION OF CROCHET WARP KNITTING MACHINE

1. The Inlay: Whilst the needle is withdrawn into its trick during knock-over of the previous warp overlap, the weft is laid below the level of the needle and on top of the warp thread that extends from its head to the warp guide.



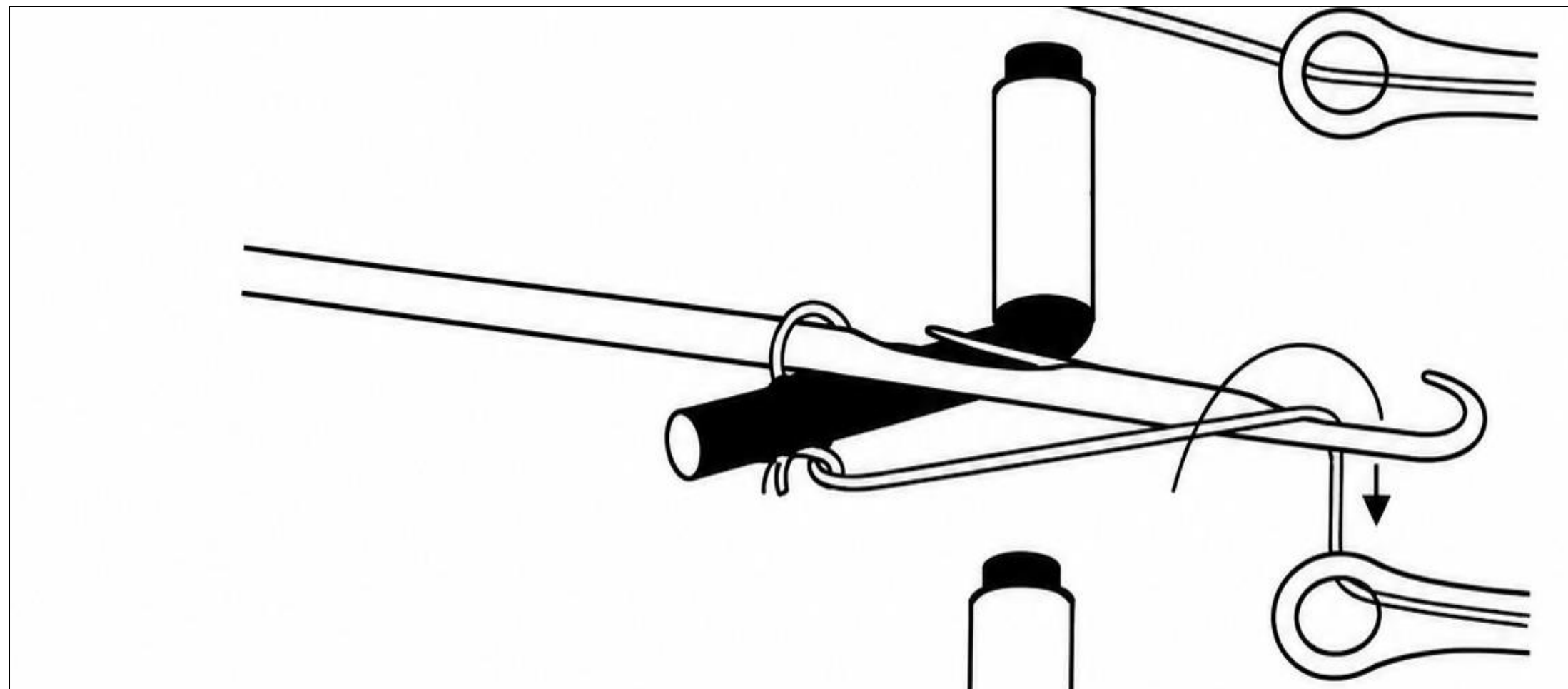
KNITTING ACTION OF CROCHET WARP KNITTING MACHINE

2. Clearing the warp overlap: The weft tube rises slightly on completion of its traverse movement to allow the needle to move out of its trick to clear its old warp overlap.



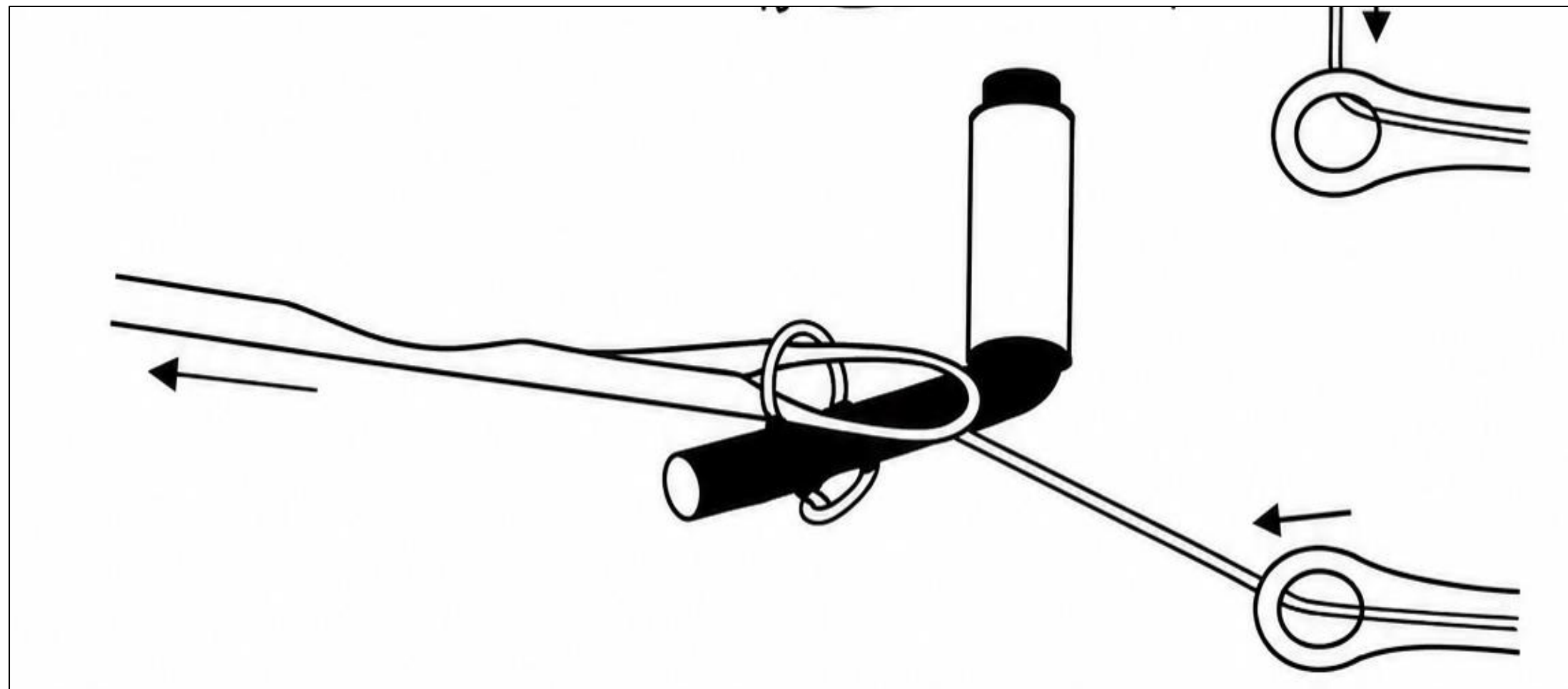
KNITTING ACTION OF CROCHET WARP KNITTING MACHINE

3. The warp overlap warp: The guide rises between the needles and automatically overlaps from the left, lowering itself again on the right side of its needle.

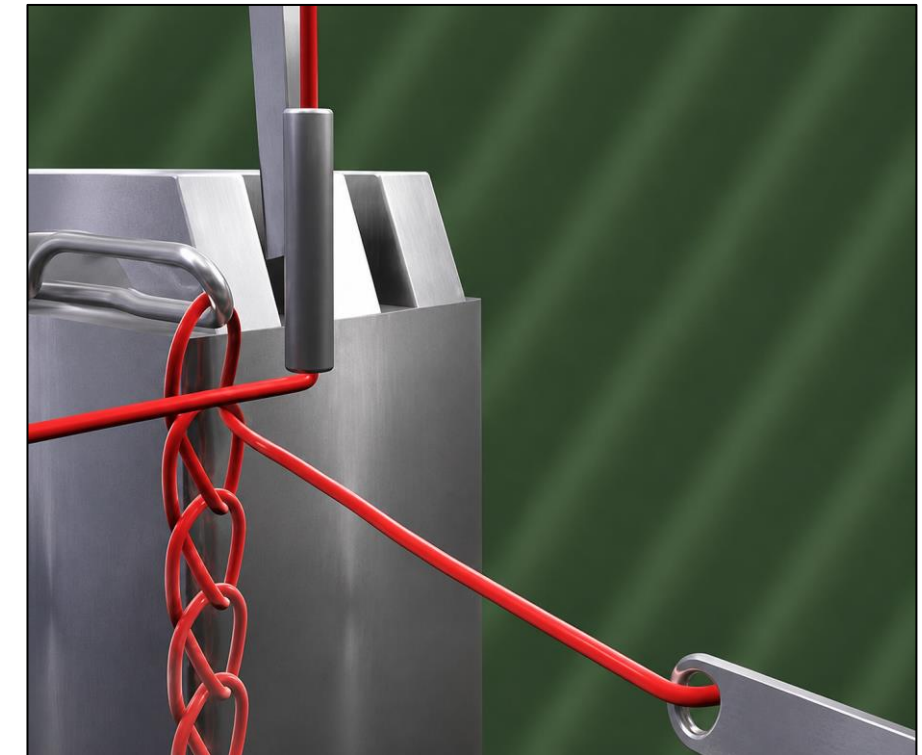
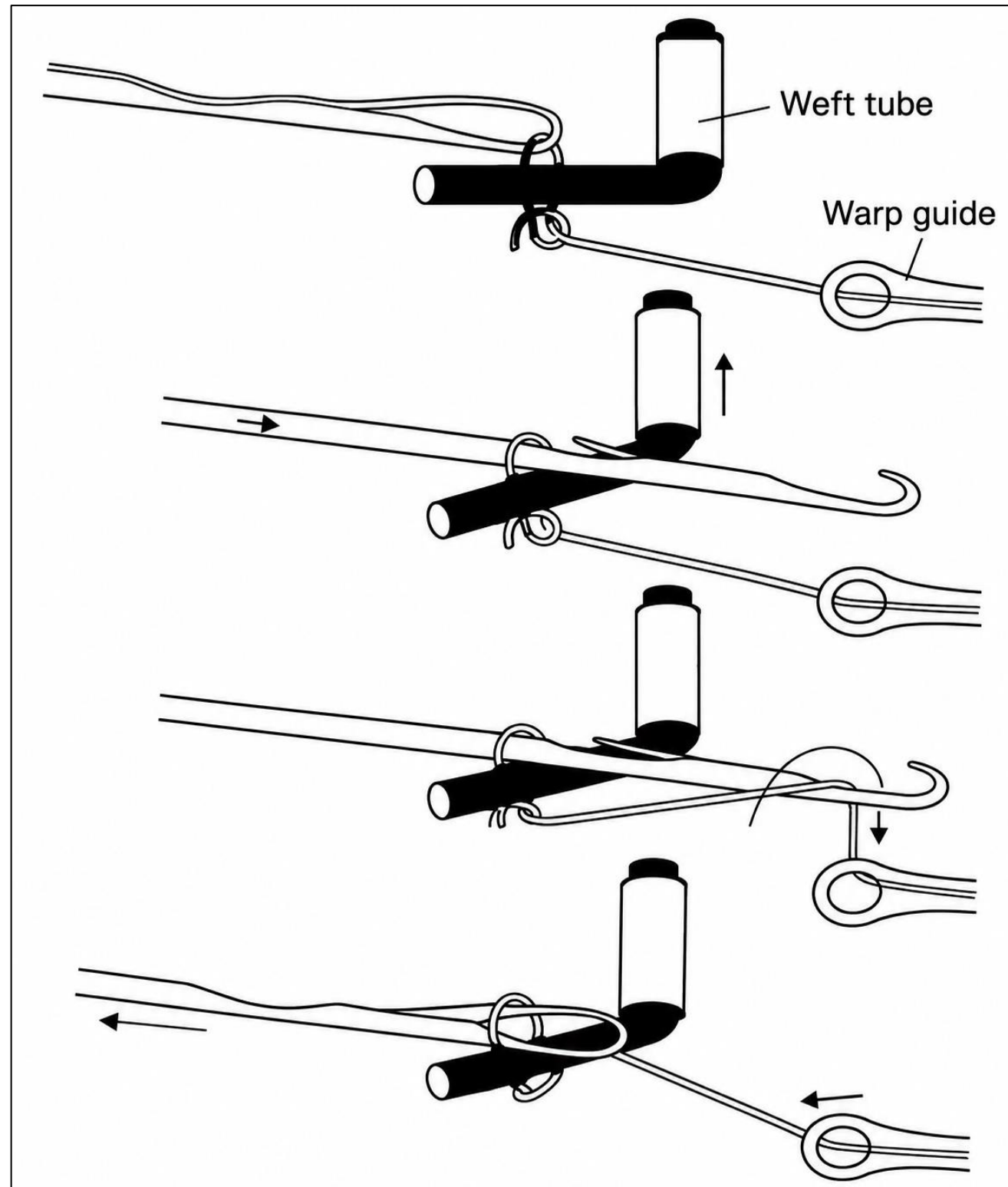


KNITTING ACTION OF CROCHET WARP KNITTING MACHINE

4. Warp knock-over and underlaps: The needle now retires into its trick to knock over the overlap, whilst the warp guide is come under its needle to the start position for its next overlap, thus completing the closed lap pillar.



KNITTING ACTION OF CROCHET WARP KNITTING MACHINE



THANK
YOU !

